

Received 6 April 2026, accepted 6 May 2026, date of publication 14 May 2026, date of current version 19 May 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3693356

RESEARCH ARTICLE

Ordered Tree-Sliced Wasserstein Distance for Sequential Data

TUNG DOAN¹, HOANG LE¹, AND HONG VUONG²

¹School of Information and Communication Technology, Hanoi University of Science and Technology, Hanoi 11615, Vietnam

²Faculty of Information Technology, VNU University of Engineering and Technology (VNU-UET), Hanoi 10000, Vietnam

Corresponding author: Tung Doan (tungdp@soict.hust.edu.vn)

This research is funded by Hanoi University of Science and Technology (HUST) under project number T2024-TD-025.

ABSTRACT Existing distance measures for sequential data are often ineffective and inefficient, as they are not true metrics and incur quadratic time complexities with respect to sequence length. In this paper, we propose a novel distance measure, called the *Ordered Tree-Sliced Wasserstein* (OTSW) distance, for sequential data. OTSW is equipped with a tree-sampling algorithm in which elements of the sequences are located at the tree nodes and structure of the tree is determined by a clustering algorithm that partitions elements based on their positional differences. Theoretical analysis demonstrates that OTSW is a valid metric for sequences and achieves linear time complexity w.r.t sequence length. We further investigate the applicability of OTSW in three tasks, including time-series classification, clustering, and visualization. Extensive experiments on widely used public datasets validate the effectiveness and efficiency of OTSW compared to existing distance measures. Our Python implementation is available at <https://github.com/TungDP/OTSW>.

INDEX TERMS Optimal transport, tree metric, sliced Wasserstein distance, sequential data, distance-based time-series classification, distance-based time-series clustering, time-series visualization.

I. INTRODUCTION

Sequential data are ubiquitous across various domains, including computer vision [1], [2], [3], [4], [5], [6], finance [7], [8], bioinformatics [9], [10], and traffic monitoring [11], [12], [13], [14], [15]. Many tasks involving sequential data rely on an effective and efficient metric for comparing sequences. However, defining such a metric is challenging for several reasons. First, sequences may vary in length, making traditional distance measures designed for fixed-size vectors unsuitable. Second, sequence elements possess an inherent order, and ignoring this ordering can lead to erroneous comparisons. Third, the complex structure of sequential data makes it difficult for a distance measure to satisfy all metric axioms, particularly the triangle inequality. Finally, sequences can be long, necessitating distance computations that are both efficient and scalable as sequence length increases.

The associate editor coordinating the review of this manuscript and approving it for publication was Juan Wang¹.

Dynamic Time Warping (DTW) [16] is arguably the most commonly used distance measure for sequential data. It aligns two sequences using a dynamic programming algorithm and computes their distance as the cumulative sum of differences between aligned samples, measured by a ground metric (e.g., Euclidean distance). Although DTW can handle sequences of varying lengths, it has three major drawbacks that limit its applicability. First, it enforces strict boundary and monotonicity constraints during alignment, which restrict its ability to handle periodic sequences with different starting points or sequences exhibiting local ordering variations [17], [18], [19]. Second, DTW is not a true metric, as it violates the triangle inequality. Finally, its quadratic time complexity with respect to sequence length makes it computationally expensive for downstream tasks such as time-series classification [20] and clustering [21].

To address the first limitation of DTW, many distance measures have recently been developed based on the Optimal Transport (OT) framework [22]. These approaches interpret the elements of each sequence as empirical samples from an

underlying distribution and apply OT to align them. Compared to DTW, OT-based alignment is more flexible because it relaxes boundary constraints and permits local reverse matching. However, the standard OT formulation does not capture global ordering relationships within sequences. As a result, current OT-based distance measures modify the ground distance matrix either explicitly [17], [23] or implicitly [17], [18], [19], [24] to incorporate the order information. These adjustments distort the metric properties of the ground distance, particularly the triangle inequality, and therefore prevent these distance measures from being true metrics. Moreover, their computation depends on Sinkhorn's algorithm [25], [26], which has quadratic time complexity with respect to sequence length. Consequently, like DTW, existing OT-based distance measures scale poorly to large datasets.

In this paper, we propose the *Ordered Tree-Sliced Wasserstein* (OTSW), a novel distance designed to address the shortcomings of existing optimal transport (OT)-based distance measures for sequential data. We first develop a tree-sampling procedure that partitions sequence elements using a hierarchical clustering algorithm and constructs a tree whose leaf nodes correspond to the elements. To incorporate order information of sequences, positional differences between elements are considered during the clustering process. The resulting trees enable the definition of tree metrics, which serve as ground distances in the formulation of OTSW. We theoretically prove that OTSW is a valid metric for sequences, as it inherits all metric properties of the ground tree metrics. Furthermore, tree metrics admit closed-form solutions for optimal transport, allowing OTSW to achieve linear time complexity with respect to sequence length. We investigate the applicability of OTSW across three tasks: time-series classification, clustering, and visualization. Extensive experiments on widely used public datasets validate the effectiveness and efficiency of the proposed distance.

In summary, the contributions of this paper are as follows:

- Development of a tree-sampling procedure in which sequence elements are located at the tree nodes, and their order information is encoded in the structure of the sampled trees.
- Introduction of the *Ordered Tree-sliced Wasserstein* (OTSW) distance, where the ground distance matrices are constructed based on the sampled trees. Theoretical analysis demonstrates that OTSW is a valid metric and achieves linear time complexity with respect to sequence length.
- Investigation of the applicability of OTSW to three tasks, including time-series classification, clustering, and visualization.
- Conducting extensive experiments on widely used public datasets to evaluate the effectiveness and efficiency of OTSW, accompanied by an in-depth discussion of the empirical results.

The rest of this paper is structured as follows. Section II reviews existing distance measures for sequential data, with a particular focus on those developed based on the OT framework. Section III outlines the essential background needed to derive the proposed distance. In Section IV, we present a tree-sampling procedure and formally define the OTSW distance based on the sampled trees, along with supporting theoretical analysis. Section V illustrates the use of the proposed distance in three tasks: time-series classification, clustering, and visualization. Section VI provides extensive experimental results evaluating the performance of OTSW, and Section VII concludes the paper.

Notation. Throughout this paper, we denote vectors and matrices by bold lowercase and bold uppercase letters, respectively. For a matrix $\mathbf{A}$, its entry at position (i, j) is denoted by $a_{i,j}$. The notation $\mathbf{1}_n$ represents an all-ones vector of n dimensions. A probability vector of uniform distribution over n elements is denoted as $\mathbf{u}_n$. $\delta_{\mathbf{x}}$ presents Dirac mass centered at $\mathbf{x}$. The notation $\langle \mathbf{A}, \mathbf{B} \rangle_F$ represents the Frobenius inner product of two matrices $\mathbf{A}$ and $\mathbf{B}$, calculated as $\sum_{i=1}^N \sum_{j=1}^M a_{i,j} b_{i,j}$.

II. RELATED WORK

To the best of our knowledge, Dynamic Time Warping (DTW) [16] remains the most commonly used distance measure for sequential data. However, as discussed above, it has three major limitations. First, DTW enforces strict boundary and monotonicity constraints during sequence alignment. The boundary constraint requires alignments to begin with the first elements and end with the last elements of both sequences, which can lead to incorrect matches for periodic sequences with different starting points. The monotonicity constraint further requires that no element in one sequence is aligned with an element in the other sequence that appears earlier than an already aligned element. Such constraint prevents DTW from handling sequences with local ordering variations [17], [18], [19]. Second, DTW has a quadratic time complexity w.r.t sequence length. Such computational cost is impractical for several downstream tasks such as time-series classification [20] and clustering [21], which require multiple pairwise distance evaluation between sequences. Third, DTW is not a true metric because it violates the triangle inequality, which can significantly degrade its effectiveness in various applications [27], [28], [29]. Although numerous DTW variants [30], [31], [32], [33], [34], [35], [36], [37], [38], [39] have been proposed, most still inherit the first limitation due to their reliance on the same alignment constraints, and none fully resolve the remaining two issues.

To explicitly address the first limitation of DTW, several distance measures for sequential data have recently been developed based on the optimal transport (OT) framework [22]. These approaches consider sequence elements as empirical samples from an underlying distribution and apply OT to align them, thereby relaxing the strict alignment

constraints imposed by DTW. However, the standard OT formulation does not account for the ordering of elements, necessitating additional mechanisms to preserve global order relationships within sequences. In general, existing OT-based distance measures can be categorized into two main groups based on how they incorporate this order information.

The first group includes distance measures that encode order information by directly altering the ground distance matrix within the classical OT framework. For instance, Temporal Coupled Optimal Transport (TCOT) [17] scale each entry of the ground distance matrix by a factor that captures the relative positional differences between sequence elements. Auto-weighted Sequential Wasserstein (ASW) [23] enhances the ground distance by incorporating a weighted combination of positional discrepancies and differences in second-order derivatives between sequences.

The second group includes distance measures that integrate order information by imposing regularization to the transport matrix. For example, Order-Preserving Wasserstein (OPW) [17] incorporates an Inverse Difference Moment (IDM) regularization [40] and enforces a Gaussian prior on the transport matrix. Time-Adaptive Optimal Transport (TAOT) [24] introduces a time-adaptive regularization to guide the alignment of sequences encoded in the transport matrix. Partial Ordered Wasserstein (POW) [18] uses a simplified version of the IDM regularization. To capture both linear and nonlinear relationships between sequences, Generalized Ordered Wasserstein (GOW) [19] defines a regularization term using a set of monotonic basis functions. Although these methods impose regularization only on the transport matrix, such regularizations implicitly alter the ground distance matrix (see A).

According to Proposition 2.3 in [22], a distance defined within the optimal transport (OT) framework is a true metric if and only if its ground distance matrix satisfies all metric axioms. However, existing OT-based distance measures modify the ground distance matrix, either explicitly or implicitly, to incorporate order information from sequences. Such modifications alter the metric properties of the ground distance matrix, leading to violations of the metric axioms and thereby preventing these distance measures from being true metrics. In contrast, our approach encodes sequence order information into tree structures via a tree-sampling procedure. These sampled trees induce tree metrics, which are used to define the ground distance matrices in the formulation of the proposed distance, called *Ordered Tree-Sliced Wasserstein* (OTSW) distance. Consequently, based on the theoretical results in [22], OTSW constitutes a valid metric for sequences. In addition, existing OT-based distance measures typically rely on Sinkhorn's algorithm [25], [26] to compute the optimal transport matrix, resulting in quadratic time complexity with respect to sequence length. In contrast, OTSW admits a closed-form solution to the OT problem

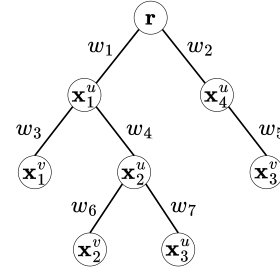


FIGURE 1. An illustration for a tree with root r and elements in $\mathcal{X}_u = \{x_1^u, x_2^u, x_3^u, x_4^u\}$ and $\mathcal{X}_v = \{x_1^v, x_2^v, x_3^v\}$ are located in its nodes.

under tree metrics, thereby achieving linear time complexity with respect to sequence length.

III. BACKGROUND

A. OPTIMAL TRANSPORT

Given two sets of elements $\mathcal{X}_u = \{x_i^u\}_{i=1}^{n_u}$ and $\mathcal{X}_v = \{x_j^v\}_{j=1}^{n_v}$ with corresponding empirical probabilities $\mu_u = \sum_{i=1}^{n_u} \alpha_i \delta_{x_i^u}$ and $\mu_v = \sum_{j=1}^{n_v} \beta_j \delta_{x_j^v}$, respectively, the Kantorovich optimal transport (OT) [22] between μ_u and μ_v is defined as follows:

$$W(\mu_u, \mu_v) = \min_{T \in \Pi(\alpha, \beta)} \langle D, T \rangle_F, \quad (1)$$

where $D \in \mathbb{R}^{n_u \times n_v}$ is the ground distance matrix, whose entry $d_{i,j} = d(x_i^u, x_j^v)$ presents the difference between x_i^u and x_j^v measured by a ground metric $d(\cdot, \cdot)$, and $T \in \mathbb{R}^{n_u \times n_v}$ is a transport matrix, which belongs to $\Pi(\alpha, \beta) = \{T \in \mathbb{R}_+^{n_u \times n_v} | T \mathbf{1}_{n_v} = \alpha, T^\top \mathbf{1}_{n_u} = \beta\}$. The vectors α and β are formed by gather all the masses α_i and β_j , respectively. In general, without any further assumptions on the prior knowledge of the elements, α and β are often set to uniform distribution vectors.

The minimum $W(\mu_u, \mu_v)$ is commonly referred to as the Wasserstein distance [22] between the two empirical probabilities μ_u and μ_v . Since μ_u and μ_v are assumed to be uniform distributions, we denote $W(\mathcal{X}_u, \mathcal{X}_v)$ as the Wasserstein distance between the two sets $\mathcal{X}_u$ and $\mathcal{X}_v$. The entry $t_{i,j}$ in the matrix T represents the amount of mass transported from x_i^u to x_j^v . In the other perspective, T can be interpreted as a soft alignment matrix, where $t_{i,j}$ indicates the degree of correspondence between x_i^u and x_j^v .

B. TREE-WASSERSTEIN DISTANCE

Problem (1) is a special case of linear programming, whose worst-case time complexity is $O(\bar{n}^3 \log \bar{n})$ with $\bar{n} = \frac{1}{2}(n_u + n_v)$. To avoid such expensive computation cost, recently, [41] and [42] leveraged tree metrics to propose tree variants of the Wasserstein distance, whose computational complexities are only $O(\bar{n})$.

Tree metrics. Given a tree $\mathcal{T}$ with non-negative, where edge weights are nonnegative and nodes correspond to elements of $\mathcal{X}_u$ and $\mathcal{X}_v$, the tree distance [43] between two elements x_i^u and x_j^v is denoted by $d_{\mathcal{T}}(x_i^u, x_j^v)$. It is defined

as the total weights of the edges along the (unique) path connecting $\mathbf{x}_i^u$ and $\mathbf{x}_j^v$. Figure 1 illustrates an example of tree, where elements in $\mathcal{X}_u = \{\mathbf{x}_1^u, \mathbf{x}_2^u, \mathbf{x}_3^u, \mathbf{x}_4^u\}$ and $\mathcal{X}_v = \{\mathbf{x}_1^v, \mathbf{x}_2^v, \mathbf{x}_3^v\}$ are located in nodes. According to this tree, the tree distance between $\mathbf{x}_2^u$ and $\mathbf{x}_3^v$, for example, is calculated as $d_{\mathcal{T}}(\mathbf{x}_2^u, \mathbf{x}_3^v) = w_4 + w_1 + w_2 + w_5$.

Tree-Wasserstein distance. Given a tree $\mathcal{T}$ with the tree distance $d_{\mathcal{T}}$, the tree-Wasserstein distance [41], [42] between two sets $\mathcal{X}_u$ and $\mathcal{X}_v$ is defined as

$$W_{\mathcal{T}}(\mathcal{X}_u, \mathcal{X}_v) = \min_{T \in \Pi(\mathbf{u}_{n_u}, \mathbf{u}_{n_v})} \langle \mathbf{D}_{\mathcal{T}}, T \rangle_F, \quad (2)$$

where $\mathbf{D}_{\mathcal{T}} \in \mathbb{R}^{n_u \times n_v}$ is the ground distance matrix, whose entry at position (i, j) is measured by $d_{\mathcal{T}}(\mathbf{x}_i^u, \mathbf{x}_j^v)$. The tree-Wasserstein distance in (2) has a following closed-form solution

$$W_{\mathcal{T}}(\mathcal{X}_u, \mathcal{X}_v) = \sum_{e \in \mathcal{T}} w_e |\mathbf{u}_{n_u}(\mathcal{T}_e) - \mathbf{u}_{n_v}(\mathcal{T}_e)|, \quad (3)$$

where w_e denotes the weight of edge e in $\mathcal{T}$, and $\mathbf{u}_{n_u}(\mathcal{T}_e)$ and $\mathbf{u}_{n_v}(\mathcal{T}_e)$ are total masses of distribution $\mathbf{u}_{n_u}$ and $\mathbf{u}_{n_v}$, respectively, within the subtree $\mathcal{T}_e$ rooted at the deeper node of the edge e . Equation (3) requires summing over all edges in the tree. In a rooted tree, the number of edges is equal to the number of nodes [42], which is bounded by $O(n_u + n_v)$. Therefore, the computational complexity of the tree-Wasserstein distance is $O(\bar{n})$.

C. TREE-SLICED WASSERSTEIN DISTANCE

In general, space of possible trees is prohibitively large. Therefore, various methods [41], [44], [45], [46], [47] have been proposed to sampled trees. Let $\{\mathcal{T}_i\}_{i=1}^t$ denotes the set of sampled trees. The tree-sliced Wasserstein distance [41] is then defined as follows:

$$TSW(\mathcal{X}_u, \mathcal{X}_v) = \frac{1}{t} \sum_{i=1}^t W_{\mathcal{T}_i}(\mathcal{X}_u, \mathcal{X}_v) \quad (4)$$

It is worth noting that computation of $W_{\mathcal{T}_i}(\mathcal{X}_u, \mathcal{X}_v)$ for each $\mathcal{T}_i$ is independent and can be performed in parallel. Therefore, the overall computational complexity of the tree-sliced Wasserstein distance remains $O(\bar{n})$.

IV. ORDERED TREE-SLICED WASSERSTEIN DISTANCE

In this section, we first present a tree-sampling procedure that incorporate order information within sequences into structures of sampled trees. We then define a novel distance, called *Ordered Tree-sliced Wasserstein* (OTSW), based tree metrics induced by these sampled trees. Finally, we provide a theoretical analysis on the properties of OTSW.

A. ORDERED TREE-SAMPLING PROCEDURE

Given a set of m sequences $\mathcal{X} = \{\mathcal{X}_u\}_{u=1}^m$, each of which is denoted by $\mathcal{X}_u = \{\mathbf{x}_i^u\}_{i=1}^{n_u}$, where n_u denotes length of the u^{th} sequence. We gather all elements from the m sequences to form a set $\mathcal{Z} = \{\mathbf{z}_p\}_{p=1}^l$, where $l = \sum_{u=1}^m n_u = m \times \bar{n}$ and $\bar{n}$ denotes the average length of sequences in $\mathcal{X}$.

Algorithm 1 Recursive Procedure for Sampling Ordered Tree

```

1: Sample_Ordered_Tree ( $\mathcal{Z}, r, n_p, h, H_{\mathcal{T}}$ )
   /*  $\mathcal{Z}$  : a set of  $l$  elements,  $r$ : the radius of  $\mathcal{Z}$ ,  $n_p$  : a parent
   node,  $h$  : a current depth level,  $H_{\mathcal{T}}$  : the prespecified
   deepest level of tree  $\mathcal{T}$  */
2: if  $l > 1$  and  $h < H_{\mathcal{T}}$  then
3:    $\mathcal{Z}_1, \mathcal{Z}_2, \dots, \mathcal{Z}_{\kappa} \leftarrow \text{FP\_Clustering}(\mathcal{Z}, \kappa)$ 
4:   for  $c = 1, \dots, \kappa$  do
5:     Insert a node  $n_c$  to  $\mathcal{T}$  as a child node of  $n_p$ .
6:      $r_c \leftarrow \frac{1}{2} \max_{\mathbf{z}_p, \mathbf{z}_q \in \mathcal{Z}_c} d_{\mathcal{T}}(\mathbf{z}_p, \mathbf{z}_q)$ .
7:      $w_{e(n_p, n_c)} \leftarrow \max(0, r - r_c)$ 
8:     Sample_Ordered_Tree ( $\mathcal{Z}_c, r_c, n_c, h+1, H_{\mathcal{T}}$ )
9:   end for
10: else
11:   Insert all elements in  $\mathcal{Z}$  to  $\mathcal{T}$  as child nodes of  $n_p$ 
12:   Set weights of all the new edges to  $\max(0, r)$ 
13:   return
14: end if

```

To sample a tree $\mathcal{T}$, in which all elements of $\mathcal{Z}$ are located at its nodes, we propose a recursive procedure, detailed in Algorithm 1. Given a parent node n_p ¹ of $\mathcal{T}$, the procedure employs farthest-point clustering [48] to partition a given set of elements $\mathcal{Z}$ into κ clusters $\mathcal{Z}_1, \mathcal{Z}_2, \dots, \mathcal{Z}_{\kappa}$ (see line 3 in Algorithm 1). For each cluster, a new node is inserted into $\mathcal{T}$ as a child of n_p . The procedure then recursively partitions each cluster into κ subclusters and performs the corresponding node insertions (see line 8 in Algorithm 1). When a subcluster contains only a single element or the tree reaches a predefined maximum depth $H_{\mathcal{T}}$, all elements of that subcluster are inserted into $\mathcal{T}$ as child nodes of the current parent node (see line 11 in Algorithm 1), after which the recursion terminates.

The farthest-point clustering algorithm [48] is summarized in Algorithm 2. Its performance depends heavily on the definition of the distance measure between elements. It is important to note that the elements in $\mathcal{Z}$ are derived from the sequences in $\mathcal{X}$. Therefore, when measuring the distance between any two elements $\mathbf{z}_p$ and $\mathbf{z}_q$, both their spatial dissimilarity and the difference in their relative positions within the corresponding sequences should be considered. Let $\mathbf{x}_i^u$ and $\mathbf{x}_j^v$ denote the elements in $\mathcal{X}$ corresponding to $\mathbf{z}_p$ and $\mathbf{z}_q$, respectively. We define the distance measure (see lines 9 and 13 in Algorithm 2) as follows:

$$d_{\mathcal{T}}(\mathbf{z}_p, \mathbf{z}_q) = \|\mathbf{x}_i^u - \mathbf{x}_j^v\|_2^2 + \tau \left(\frac{i}{n_u} - \frac{j}{n_v} \right)^2, \quad (5)$$

where the first term captures spatial dissimilarity, and the second term quantifies the relative difference in ordering between the two elements within their respective sequences. The parameter $\tau > 0$ balances the contributions of these two terms.

¹ At the first time of calling the procedure, the parent node n_p corresponds to the root of the tree.

Algorithm 2 Farthest Point Clustering

```

1: FP_Clustering( $\mathcal{Z}, \kappa$ )
   /*  $\mathcal{Z} = \{z_p\}_{p=1}^l$  : a set of elements,  $\kappa$  : the number of
   clusters */
2: Initialize an empty set of cluster centers  $C$ 
3: Initialize  $\kappa$  empty clusters  $\mathcal{Z}_1, \mathcal{Z}_2, \dots, \mathcal{Z}_\kappa$ 
   /* select cluster centers */
4:  $y_1 \leftarrow$  a random element in  $\mathcal{Z}$ 
5:  $C \leftarrow C \cup y_1$ 
6:  $c \leftarrow 1$ 
7: while  $c < \kappa$  and  $l - c > 0$  do
8:    $c \leftarrow c + 1$ 
9:    $y_c \leftarrow \max_{z \in \mathcal{Z}} \min_{y \in C} d_\tau(z, y)$ 
10:   $C \leftarrow C \cup y_c$ 
11: end while
   /* partition  $\mathcal{Z}$  into  $\kappa$  clusters */
12: for  $p = 1, 2, \dots, l$  do
13:    $c \leftarrow \operatorname{armin}_{k=1, \dots, \kappa} d_\tau(z_p, y_k)$ 
14:    $\mathcal{Z}_c \leftarrow \mathcal{Z}_c \cup z_p$ 
15: end for
16: return  $\mathcal{Z}_1, \mathcal{Z}_2, \dots, \mathcal{Z}_\kappa$ 

```

Besides determining structure of the sampled tree, Algorithm 1 also assigns a weight to each edge in $\mathcal{T}$. Specifically, each node in the tree represents a cluster.² We define the radius of a cluster as the largest distance d_τ between any two of its elements (see line 6 in Algorithm 1). Note that clusters represented by leaf nodes contain only a single element and therefore have zero radius. Given an edge connecting a parent node n_p and a child node n_c , let r and r_c denote their respective radii. The weight of this edge (see line 7 in Algorithm 1) is then computed as

$$w_{e(n_p, n_c)} \leftarrow \max(0, r - r_c) \quad (6)$$

It is worth noting that the weight of an edge connecting a parent node to a leaf node simplifies to $\max(0, r)$, since leaf nodes have zero radius (see line 12 in Algorithm 1).

B. DEFINITION

Due to the randomness of the farthest-point clustering algorithm in selecting cluster centers, recursive procedure detailed in Algorithm 1 allows us to sample various trees. These trees encode order information of the sequential elements in their structures and weights. Therefore, we call them *ordered trees*. In practice, we use the recursive procedure to sample t ordered trees $\{\mathcal{T}_i\}_{i=1}^t$. These trees have all elements in $\mathcal{X}$ located on their leaf nodes. Therefore, when comparing two sequences $\mathcal{X}_u$ and $\mathcal{X}_v$ from $\mathcal{X}$, we may remove all edges and leaf nodes that are not related to these sequences from $\{\mathcal{T}_i\}_{i=1}^t$, obtaining simplified versions of trees, denoted as $\{\mathcal{T}_i^{u,v}\}_{i=1}^t$. Then, we define the *Ordered*

²The root represents the entire set of elements $\mathcal{Z}$, whereas each leaf node corresponds to an individual element in $\mathcal{Z}$.

Tree-sliced Wasserstein (OTSW) distance between $\mathcal{X}_u$ and $\mathcal{X}_v$ as follows:

$$OTSW(\mathcal{X}_u, \mathcal{X}_v) = \frac{1}{t} \sum_{i=1}^t W_{\mathcal{T}_i^{u,v}}(\mathcal{X}_u, \mathcal{X}_v), \quad (7)$$

where $W_{\mathcal{T}_i^{u,v}}(\mathcal{X}_u, \mathcal{X}_v)$ is the *Ordered Tree Wasserstein* (OTW) distance between $\mathcal{X}_u$ and $\mathcal{X}_v$, defined as:

$$W_{\mathcal{T}_i^{u,v}}(\mathcal{X}_u, \mathcal{X}_v) = \min_{T \in \Pi(u_{n_u}, u_{n_v})} \langle \mathcal{D}_{\mathcal{T}_i^{u,v}}, T \rangle_F. \quad (8)$$

Because OTW is also a tree-Wasserstein distance, it has a closed-form solution as in (3).

C. THEORETICAL ANALYSIS

Complexity. Prior to distance computation, Algorithm 1 is used to sample t ordered trees for the dataset $\mathcal{X} = \{\mathcal{X}_u\}_{u=1}^m$, which contains $m \times \bar{n}$ elements. Notably, the algorithm can be executed in parallel to generate these trees; therefore, the total computational cost of tree sampling is equivalent to that of a single execution of Algorithm 1. The core computation at each recursive step of the algorithm involves partitioning a set of elements into κ clusters using the farthest-point clustering algorithm [48]. In general, this clustering algorithm partitions n elements into κ clusters in $O(n\kappa)$ time. In practice, an accelerated implementation [49] may be adopted, achieving $O(n \log \kappa)$ complexity. Since Algorithm 1 starts with $m \times \bar{n}$ elements and κ is typically much smaller than $m \times \bar{n}$, the Master Theorem [50] implies an overall time complexity of $O(m\bar{n}(\log m + \log \bar{n}))$ for sampling an ordered tree.

Given t ordered trees $\{\mathcal{T}_i\}_{i=1}^t$, computing OTSW distance between two sequences $\mathcal{X}_u$ and $\mathcal{X}_v$ of lengths n_u and n_v only requires the simplified trees $\{\mathcal{T}_i^{u,v}\}_{i=1}^t$, obtained by removing nodes and edges unrelated to $\mathcal{X}_u$ and $\mathcal{X}_v$. Since each simplified tree has the number of nodes bounded by $O(n_u + n_v)$, time for computing each OTW distance (8) in closed form is linear w.r.t the average of sequence lengths. The OTSW distance (7) is then computed as an average over the OTW distances. It is worth noting that computations of OTW distances can also be parallelized. Therefore, the overall time complexity of OTSW is $O(\bar{n})$.

Metric properties. We can show the following proposition that clarifies metric properties of OTSW distance:

Proposition 1: The OTSW distance defined in (7) satisfies all axioms of a metric:

- Non-negativity: $OTSW(\mathcal{X}_u, \mathcal{X}_v) \geq 0$ for all $\mathcal{X}_u, \mathcal{X}_v$.
- Identity: $OTSW(\mathcal{X}_u, \mathcal{X}_v) = 0 \Leftrightarrow \mathcal{X}_u = \mathcal{X}_v$.
- Symmetry: $OTSW(\mathcal{X}_u, \mathcal{X}_v) = OTSW(\mathcal{X}_v, \mathcal{X}_u)$ for all $\mathcal{X}_u, \mathcal{X}_v$.
- Triangle inequality: $OTSW(\mathcal{X}_u, \mathcal{X}_v) \leq OTSW(\mathcal{X}_u, \mathcal{X}_w) + OTSW(\mathcal{X}_w, \mathcal{X}_v)$ for all $\mathcal{X}_u, \mathcal{X}_v, \mathcal{X}_w$.

Proof: Ordered trees sampled using Algorithm 1 have nonnegative edge weights (see Equation (6)), and each pair of its nodes is connected by a unique path [48]. Consequently, the tree distances defined on these ordered trees constitute tree metrics and, as noted in [51], satisfy all metric axioms,

including nonnegativity, identity, symmetry, and the triangle inequality.

These tree distances are then used to construct the ground distance matrix for the OTW distance defined in Equation (8). Since the OTW formulation parallels that of the Wasserstein distance (see Equation (1)), Proposition 2.3 in [22] implies that the OTW distance also satisfies all metric axioms.

Finally, because the OTSW distance is defined as the average of OTW distances (see Equation (7)), it inherits all metric properties from the OTW distance and is therefore a valid metric for sequences. ■

V. APPLICATIONS

In this work, we propose a new distance for sequential data. It is widely adopted in the literature [17], [18], [19], [24] that downstream tasks provide a practical way to evaluate both the effectiveness and efficiency of distance measures. Accordingly, in this section, we investigate the application of the OTSW distance to three tasks: time-series classification, clustering, and visualization. For each task, we begin by defining the problem, then explain how OTSW can be utilized to solve it, and finally analyze the associated computational complexity.

A. TIME-SERIES CLASSIFICATION

Problem. Time-series classification (TSC) is a fundamental task in time-series analysis due to its broad range of applications, including human action recognition [52], [53], [54], fault detection [57], [58], and electrocardiogram diagnosis [55], [56], to name a few. The objective of TSC is to assign the correct class label to an unlabeled time series by leveraging a set of labeled sequences, with the aim of achieving the highest possible prediction accuracy.

Application of the distance measure. To classify unlabeled time series, we use the k -nearest neighbors (k -NN) classifier in combination with OTSW. In the experimental section, we also pair k -NN with existing distance measures to enable a fair comparison. Our goal is to assess the quality of OTSW by examining its classification performance relative to these baselines. Due to its simplicity, the performance of k -NN provides a direct and comprehensive reflection of the effectiveness of the underlying distance measure. Furthermore, despite its straightforward nature, k -NN continues to be a widely adopted approach for time-series classification. Prior studies [17], [18], [19], [59], [60], [61] have demonstrated that combining k -NN with existing sequence distance measures often produces strong results that are difficult to significantly outperform.

Time complexity. Given a training set of m time series with an average length of $\bar{n}$, assigning a label for an unlabeled time series using the k -NN classifier requires a computational cost of $O(mT_d + m \log m)$, where T_d represents the time of distance evaluation between the query series and each training series. For commonly used distance measures such as DTW, OPW, TCOT, TAOT, ASW, POW, and GOW, a single distance

computation has quadratic complexity in the sequence length. As a result, the total complexity of k -NN with these distances is at least $O(m\bar{n}^2 + m \log m)$ for classifying a single query.

For the proposed distance, the tree-sampling procedure is performed only once, with a time complexity of $O(m\bar{n}(\log m + \log \bar{n}))$. Thereafter, assigning a label to an unlabeled time series requires $O(m\bar{n} + m \log m)$ time. The total computational cost is thus $O((\bar{n} + 1)m \log m + m\bar{n}(\log \bar{n} + 1))$, which remains asymptotically linear in the sequence length.

B. TIME-SERIES CLUSTERING

Problem. Time-series clustering (TSCL) is another fundamental task in time-series analysis, with applications spanning diverse areas such as community detection [62], signal processing [63], anomaly detection [64], [65], and human action recognition [66], [67]. The goal of TSCL is to partition a collection of time series into clusters such that those within the same cluster are highly similar, while those in different clusters are highly dissimilar.

Application of the distance measure. To cluster time series, we adopt the k -medoids algorithm [68]. Similar to the classification experiments, we incorporate OTSW and existing distance measures, into the k -medoids framework to enable a consistent comparison. Since clustering is an unsupervised task and k -medoids is a relatively simple algorithm, the quality of the resulting clusters heavily depends on the effectiveness of the chosen distance measure. Consequently, clustering performance serves as a reliable criterion for evaluating the proposed distance and benchmarking it against existing baselines. Additional details are provided in the experimental section.

Time complexity. The time complexity of the k -medoids algorithm for clustering m time series with an average length of $\bar{n}$ into k groups is $O(km^2T_d)$, where T_d denotes the cost of computing the distance between each pair of time series. For existing distance measures such as DTW, OPW, TCOT, TAOT, ASW, POW, and GOW, a single distance computation has quadratic complexity with respect to sequence length. As a result, the overall complexity of k -medoids with these measures is at least $O(km^2\bar{n}^2)$.

For the proposed distance, its time complexity is linear w.r.t to series length. When the additional cost of sampling ordered trees is taken into account, the total time complexity of k -medoids using OTSW becomes $O(m\bar{n}(km + \log m + \log \bar{n}))$, which is much more efficient than existing distance measures, particularly when m and/or $\bar{n}$ are large.

C. TIME SERIES VISUALIZATION

Problem. Interpreting time series has become an increasingly important topic across many applications [69], [70], [71], [72], particularly as technological advances give rise to rapidly growing and often high-dimensional time-series datasets. In this paper, we focus on the problem of time-series visualization, which aims to learn a low-dimensional

representation for each sequence while preserving as much essential information from the original data as possible.

Application of the distance measure. To learn a low-dimensional representation for each time series, we employ t-SNE [73], a widely used representation learning technique. Given a set of m time series $\mathcal{X} = \{\mathcal{X}_u\}_{u=1}^m$, t-SNE learns representation (e.g., two- or three-dimensional vectors) $\mathcal{Y} = \{y_u\}_{u=1}^m$ in Euclidean space by minimizing the following objective:

$$\sum_{u=1}^m \sum_{v \neq u} p_{u,v} \log \frac{p_{u,v}}{q_{u,v}}, \quad (9)$$

where

$$p_{u,v} = \frac{1}{2m} \left(\frac{e^{-\frac{d(\mathcal{X}_u, \mathcal{X}_v)^2}{2\sigma_u^2}}}{\sum_{u=1}^m \sum_{v \neq u} e^{-\frac{d(\mathcal{X}_u, \mathcal{X}_v)^2}{2\sigma_u^2}}} + \frac{e^{-\frac{d(\mathcal{X}_v, \mathcal{X}_u)^2}{2\sigma_v^2}}}{\sum_{v=1}^m \sum_{w \neq v} e^{-\frac{d(\mathcal{X}_v, \mathcal{X}_w)^2}{2\sigma_v^2}}} \right), \quad (10)$$

and

$$q_{u,v} = \begin{cases} \frac{(1 + \|y_u - y_v\|_2^2)^{-1}}{\sum_{r=1}^m \sum_{t \neq r} (1 + \|y_r - y_t\|_2^2)^{-1}} & \text{if } u \neq v, \\ 0 & \text{otherwise.} \end{cases} \quad (11)$$

In equation (10), σ_u and σ_v denote the Gaussian kernel bandwidths, which can be determined via binary search [74] or robust root-finding methods [75]. The function $d(\cdot, \cdot)$ represents a distance measure between time series. To compare OTSW with existing sequence distances, we incorporate each distance measure into t-SNE through $d(\cdot, \cdot)$ and evaluate the resulting t-SNE performance.

Time complexity. Objective (9) consists of two key components, $p_{u,v}$ and $q_{u,v}$. Computing $p_{u,v}$ requires approximately m^2 evaluations of distances between sequences, while computing $q_{u,v}$ similarly involves approximately m^2 distance evaluations between low-dimensional representations. Let T_d denote the time complexity of computing the distance between two sequences. Since distance computations between sequences are significantly more expensive than those between low-dimensional representations, the overall time complexity of evaluating the t-SNE objective is $O(m^2 T_d)$.

For existing distance measures such as DTW, OPW, TCOT, TAOT, ASW, POW, and GOW, the cost of distance computation grows quadratically with sequence length. Consequently, t-SNE integrated with these distance measures has a time complexity of $O(m^2 \bar{n}^2)$, where $\bar{n}$ denotes the average length of the time series. In contrast, the proposed OTSW distance has computational complexity that is linear in sequence length. Accounting for the cost of sampling ordered trees, the overall time complexity of t-SNE with OTSW becomes $O(m\bar{n}(m + \log m + \log \bar{n}))$.

TABLE 1. Statistics of the 20 datasets selected from the UCR Archive. For datasets with long names, an abbreviated alternative name is provided in the second column.

Dataset	Alternative name	Length	Train	Test	Classes
ArrowHead	AH	251	36	175	3
BasicMotions	BM	100	40	40	4
BeetleFly	BF	512	20	20	2
CBF	—	128	30	900	3
Chinatown	CT	24	20	343	2
CinCECGTorso	CET	1639	40	1380	4
DiatomSizeReduction	DSR	345	16	306	4
GunPointAgeSpan	GPA	150	135	316	2
GunPointMaleVersusFemale	GPM	150	135	316	2
GunPointOldVersusYoung	GPO	150	136	315	2
Ham	—	431	109	105	2
InsectEPGRegularTrain	IERT	601	62	249	3
ItalyPowerDemand	IPD	24	67	1029	2
Meat	—	448	60	60	3
MelbournePedestrian	MP	24	1194	2439	10
MixedShapesSmallTrain	MS2T	1024	100	2425	5
MoteStrain	MS	84	20	1252	2
OliveOil	O2	570	30	30	4
Plane	—	144	105	105	7
SmoothSubspace	S2	15	150	150	3



FIGURE 2. Binary features extracted from human action videos in Weizmann dataset.

VI. EXPERIMENTS

A. DATASETS

UCR time series datasets. The UCR archive [76] provides a large collection of time-series datasets covering diverse application domains such as finance, biology, industrial systems, and healthcare. From this repository, we selected 20 datasets, each consisting of equal-length univariate time series associated with distinct class labels. All datasets come with predefined training and test sets, and their key statistics are reported in Table 1. For the clustering and visualization experiments, the training and test sets are merged before performing the tasks.

Weizmann dataset. The Weizmann (WEI) dataset [77] comprises 90 video sequences featuring nine subjects, each performing ten different actions. In the preprocessing pipeline, the background is first removed from each frame, after which the frames are resized to 70×35 pixels. From each processed frame, a binary feature representation is extracted, as illustrated in Figure 2. To reduce the dimensionality of the resulting 2450-dimensional data, Principal Component Analysis (PCA) is applied, and the top 123 principal components, which capture 99% of the total variance, are retained. For clustering and visualization experiments, the entire dataset is used. In contrast, for classification, the dataset is divided into training and test sets: video sequences from the first five subjects are used for training, while those from the remaining four subjects are reserved for testing.

MSR Sport Action3D dataset. The MSR Sports Action3D (MSA) dataset [78] contains 402 depth sequences



FIGURE 3. A sequence of 3D skeleton joint coordinates representing jogging action, selected from the MSR Sport Action3D dataset.

representing 20 different sports-related actions, each performed three times by ten subjects, for a total of 23,797 frames. For each frame, we use the raw 3D skeleton joint coordinates as feature representations, which are illustrated in Figure 3. For the classification task, we adopt the evaluation protocol from [17] and [19], where sequences performed by the first half of the subjects are used for training and those from the remaining subjects are used for testing. For clustering and visualization, we perform the tasks on the entire dataset without considering the training–testing split.

Spoken Arabic Digits dataset. The Spoken Arabic Digits (SAD) dataset [79] contains recordings of 10 Arabic digits spoken by 88 individuals, with each speaker repeating every digit 10 times. For the clustering and visualization experiments, we use the entire dataset. In contrast, for the classification task, we follow the protocol in [17] and [19], selecting 660 samples per digit (for a total of 6,600 audio sequences) as the training set and using the remaining 2,200 sequences for testing. Each audio sample is represented as a sequence of feature vectors, where each vector consists of 13 mel-frequency cepstral coefficients.

B. PERFORMANCE MEASURES

Classification. For the classification tasks, we evaluate the performance of the k -NN classifier combined with different distance measures on the test sets. Following [17], [18] and [19], we use *accuracy* (ACC) and *mean average precision* (MAP) as evaluation metrics. Accuracy is defined as the percentage of correctly classified sequences in the test set. To compute MAP, we first determine the average precision (AP) for each class, which corresponds to the area under the precision–recall curve. The MAP is then calculated by averaging the AP values over all classes.

Clustering. For the clustering tasks, we evaluate the performance of the k -medoids algorithm combined with different distance measures on the full dataset. *Accuracy* (ACC) and *normalized mutual information* (NMI) are employed as evaluation metrics [80].

Let $\hat{\mathcal{C}} = \{\hat{\mathcal{C}}_u\}_{u=1}^m$ and $\mathcal{C} = \{\mathcal{C}_u\}_{u=1}^m$ denote the predicted labels and ground-truth labels, respectively, for the set $\mathcal{X} = \{\mathcal{X}_u\}_{u=1}^m$. The assignment $\hat{\mathcal{C}}_u = i$ (similarly for $\mathcal{C}_u$), for $1 \leq i \leq k$, indicates that sequence $\mathcal{X}_u$ belongs to the i -th cluster. The accuracy is defined as

$$ACC = \frac{\sum_{u=1}^m \delta(\mathcal{C}_u, \text{map}(\hat{\mathcal{C}}_u))}{m}, \quad (12)$$

where $\delta(a, b)$ denotes the indicator function, which equals one if $a = b$ and zero otherwise, and $\text{map}(\hat{\mathcal{C}}_u)$ is a permutation function that maps the predicted label $\hat{\mathcal{C}}_u$ to the corresponding ground-truth label. In this work, the optimal mapping is obtained using the Kuhn–Munkres algorithm [81].

Let $\hat{\mathcal{C}} = \{\hat{\mathcal{C}}_i\}_{i=1}^k$ and $\mathcal{C} = \{\mathcal{C}_i\}_{i=1}^k$ denote the obtained clusters and the ground-truth clusters, respectively. Their mutual information (MI) is defined as

$$MI(\mathcal{C}, \hat{\mathcal{C}}) = \sum_{\mathcal{C}_i \in \mathcal{C}, \hat{\mathcal{C}}_j \in \hat{\mathcal{C}}} p(\mathcal{C}_i, \hat{\mathcal{C}}_j) \log_2 \frac{p(\mathcal{C}_i, \hat{\mathcal{C}}_j)}{p(\mathcal{C}_i)p(\hat{\mathcal{C}}_j)}, \quad (13)$$

where $p(\mathcal{C}_i)$ and $p(\hat{\mathcal{C}}_j)$ denote the probabilities that a randomly selected sequence belongs to clusters $\mathcal{C}_i$ and $\hat{\mathcal{C}}_j$, respectively, and $p(\mathcal{C}_i, \hat{\mathcal{C}}_j)$ denotes the joint probability that a randomly selected sequence belongs to both $\mathcal{C}_i$ and $\hat{\mathcal{C}}_j$. This measure is normalized to the range $[0, 1]$ as

$$NMI(\mathcal{C}, \hat{\mathcal{C}}) = \frac{MI(\mathcal{C}, \hat{\mathcal{C}})}{\max(H(\mathcal{C}), H(\hat{\mathcal{C}}))}, \quad (14)$$

where $H(\mathcal{C})$ and $H(\hat{\mathcal{C}})$ denote the entropies of $\mathcal{C}$ and $\hat{\mathcal{C}}$, respectively.

Visualization. In the visualization task, we learn a representation vector for each sequence. To quantify quality of the learned vectors, we employ three evaluation metrics, including *trustworthiness* [82], *continuity* [83], and the *local continuity meta-criterion* (LCMC) [85].

Let $\mathcal{H}$ denote the space of the given m sequences, and let $\mathcal{L}$ denote the space of their corresponding low-dimensional representation vectors. The rank of the closeness between sequence u and sequence v in $\mathcal{H}$ is denoted by $r^{\mathcal{H}}(u, v)$, where $r^{\mathcal{H}}(u, v) = 1$ if sequence v is the nearest neighbor of sequence u . The rank of the closeness between the corresponding representation vectors in $\mathcal{L}$ is denoted by $r^{\mathcal{L}}(u, v)$, defined analogously. We further denote by $N_k^{\mathcal{H}}(u)$ and $N_k^{\mathcal{L}}(u)$ the sets of the k nearest neighbors of u in $\mathcal{H}$ and $\mathcal{L}$, respectively.

The *trustworthiness* between $\mathcal{H}$ and $\mathcal{L}$ is then defined as

$$T(k) = 1 - \frac{2}{mk(2m - 3k - 1)} \sum_{u=1}^m \sum_{v \in P_k(u)} (r^{\mathcal{H}}(u, v) - k), \quad (15)$$

where $P_k(u) = N_k^{\mathcal{L}}(u) \setminus N_k^{\mathcal{H}}(u)$ is the set of sequences that are not neighbors of sequence u in $\mathcal{H}$, but whose representation vectors are neighbors of that of sequence u in $\mathcal{L}$.

The *continuity* between $\mathcal{H}$ and $\mathcal{L}$ is defined as

$$C(k) = 1 - \frac{2}{mk(2m - 3k - 1)} \sum_{u=1}^m \sum_{v \in Q_k(u)} (r^{\mathcal{L}}(u, v) - k), \quad (16)$$

where $Q_k(u) = N_k^{\mathcal{H}}(u) \setminus N_k^{\mathcal{L}}(u)$ denotes the set of sequences that are neighbors of sequence u in $\mathcal{H}$, but whose corresponding representation vectors are not neighbors of that of sequence u in $\mathcal{L}$.

The *local continuity meta-criterion* between $\mathcal{H}$ and $\mathcal{L}$ is defined as

$$LCMC(k) = \text{Overlap}(k) - \frac{k}{m-1}, \quad (17)$$

where

$$\text{Overlap}(k) = \frac{1}{mk} \sum_{u=1}^m |N_k^{\mathcal{H}}(u) \cap N_k^{\mathcal{L}}(u)|. \quad (18)$$

The three metrics described above depend on the parameter k , which specifies the number of considered neighbors. Following prior studies [82], [83], [84], we varied parameter k from 4 to 20, and found that our proposed distance and baselines achieve the highest scores at $k = 5$.

C. BASELINES

We compare OTSW with the following baselines (see Sections II):

- Dynamic time warping (DTW) [16] - a popular distance measure for sequences that impose boundary, monotonicity and continuity constraints on sequence alignment.
- Order preserving Wasserstein (OPW) [17] - an OT-based distance measure that imposes inverse difference moment (IDM) regularization and Gaussian prior on the transport matrix to preserve order information of sequences.
- Temporally coupled optimal transport (TCOT) [17] - an OT-based distance measure that directly integrates positional difference between sequence elements into the ground distance matrix.
- Time adaptive optimal transport (TAOT) [24] - an OT-based distance measure that imposes time adaptive regularization on the transport matrix to preserve the order information of sequences.
- Auto-weighted sequential Wasserstein (ASW) [23] - an OT-based distance measure that defines the ground distance matrix as a weighted sum of pairwise spatial, structural and order distance matrices.
- Partial ordered Wasserstein (POW) [18] - an OT-based distance measure that uses a simplified version of IDM regularization and automatically limits the transportation between sequences.
- Generalized ordered Wasserstein (GOW) [19] - an OT-based distance measure that imposes generalized ordered regularization on the transport matrix to preserve the order information of sequences.

D. PARAMETER TUNING

For the classification task, we employ the k -nearest neighbors (k -NN) classifier, where the number of neighbors k is a critical hyperparameter. In all classification experiments, k is selected from the set $\{1, 3, 5, 7, 15, 30\}$. We observe that choosing $k = 1$ consistently yields high classification accuracy across most distance measures. For the clustering task, the number of clusters must be specified in advance

for the k -medoids algorithm. Since our goal is to fairly evaluate performances of different distance measures, we set the clustering parameter k equal to the ground-truth number of clusters for each dataset. For the visualization task, we consider three important hyperparameters of t-SNE: the Gaussian kernel bandwidths σ_u and σ_v (see Equation (10)), and the learning rate that controls the step size of gradient descent in minimizing the t-SNE objective. The Gaussian kernel bandwidths are determined using a root-finding method [75]. The learning rate is selected from the set $\{0.01, 0.1, 1, 10, 100, 1000\}$, and t-SNE is observed to achieve consistently strong performance across all datasets when the learning rate lies in the range $[0.1, 10]$.

Our proposed distance involves four key hyperparameters. The parameter τ (see Equation 5) balances the spatial dissimilarity and the relative ordering difference between two sequence elements. The parameter κ (see Algorithm 2) denotes the number of clusters used when selecting ordered trees. The parameter H_T (see Algorithm 1) specifies the maximum depth of the selected ordered trees. Finally, t represents the number of ordered trees used to construct OTSW (see Equation 7). We employ grid search to select optimal values for these hyperparameters across different datasets. Specifically, τ is selected from $\{0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1, 5, 10, 50, 100, 10^3, 10^4\}$; κ is chosen from $\{2^a\}_{a=1}^{12}$; H and t are selected from $\{5, 10, 15, 20, 25, 30\}$ and $\{1, 3, 5, 7, 9, 11, 13, 15\}$, respectively. The sensitivity of the proposed distance to these hyperparameters is discussed in detail in the ablation study.

All baseline distance measures involve ground distance matrices that present pairwise distance between sequence elements. In our experiments, we use the Euclidean distance as the ground metric. The remaining hyperparameters of these baselines are set according to the configurations reported in their respective original papers.

E. RESULTS AND DISCUSSION

Classification. We first evaluate the performance of OTSW when integrated with the k -NN classifier for the time-series classification task. Table 2 reports the MAP and ACC scores of k -NN classifiers equipped with different distance measures. The results show that OTSW enables the k -NN classifier to achieve the highest performance across both evaluation metrics on 16 out of the 23 datasets. On an additional 5 datasets, OTSW outperforms all baseline methods in terms of either MAP or ACC.

To further strengthen the evaluation, we conduct McNemar's test [86] to compare OTSW with the best-performing baseline. The results are also reported in Table 2. We can observe that OTSW significantly outperforms the best-performing baselines on 6 datasets, regardless of evaluation metric. For an additional 7 datasets, OTSW significantly improves the performance of the k -NN classifier compared with the best baselines in terms of either MAP or ACC. In most of the remaining cases, except for the

TABLE 2. Classification results of the k -NN classifier combined with different distance measures. For each dataset, the classifiers were executed five times and the average scores along with standard deviation are reported. The top three highest scores for each dataset are presented in bold, italic red and underlined blue, respectively. The symbol $\dagger$ indicates that the performance of the proposed distance is not significantly different from the best performance from baselines at the $\alpha = 5\%$ level, using McNemar's test [86], while the symbol $*$ indicates that it is significantly different.

Datasets		DTW	OPW	TCOT	TAOT	ASW	POW	GOW	OTSW
AH	MAP	65.64 (0.00)	64.39 (0.00)	59.50 (0.00)	<u>75.01 (0.00)</u>	65.90 (0.00)	63.77 (0.00)	<i>77.15 (0.00)</i>	79.80 (0.07)*
	ACC	70.86 (0.00)	67.43 (0.00)	67.43 (0.00)	<u>76.57 (0.00)</u>	74.86 (0.00)	73.14 (0.00)	<i>78.21 (0.00)</i>	80.00 (0.05)*
BM	MAP	94.07 (0.00)	72.21 (0.00)	63.22 (0.00)	<i>97.86 (0.00)</i>	70.05 (0.00)	74.15 (0.00)	99.15 (0.00)	<u>95.23 (0.02)*</u>
	ACC	<i>97.50 (0.00)</i>	75.00 (0.00)	70.50 (0.00)	76.15 (0.00)	75.00 (0.00)	76.50 (0.00)	100.00 (0.00)	<u>92.50 (0.00)*</u>
BF	MAP	66.25 (0.00)	60.50 (0.00)	79.87 (0.00)	92.95 (0.00)	<u>93.61 (0.00)</u>	85.15 (0.00)	<i>94.37 (0.00)</i>	95.56 (0.03)*
	ACC	70.00 (0.00)	65.00 (0.00)	<u>85.00 (0.00)</u>	70.00 (0.00)	<u>90.00 (0.00)</u>	<i>90.00 (0.00)</i>	95.00 (0.00)	95.00 (0.00)†
CBF	MAP	99.45 (0.00)	97.12 (0.00)	96.91 (0.00)	91.70 (0.00)	<i>99.63 (0.00)</i>	<i>99.63 (0.00)</i>	100.00 (0.00)	<u>99.50 (0.03)†</u>
	ACC	99.67 (0.00)	98.22 (0.00)	98.11 (0.00)	94.56 (0.00)	<i>99.78 (0.00)</i>	<i>99.78 (0.00)</i>	<u>99.75 (0.00)</u>	99.86 (0.01)†
CT	MAP	<i>95.06 (0.00)</i>	64.42 (0.00)	57.08 (0.00)	78.85 (0.00)	93.12 (0.00)	89.00 (0.00)	<u>95.00 (0.00)</u>	96.67 (0.05)†
	ACC	<u>97.38 (0.00)</u>	77.84 (0.00)	54.23 (0.00)	87.76 (0.00)	93.29 (0.00)	94.17 (0.00)	<i>97.50 (0.00)</i>	98.45 (0.03)*
CET	MAP	51.76 (0.00)	48.12 (0.00)	45.33 (0.00)	<u>68.72 (0.00)</u>	44.62 (0.00)	42.15 (0.00)	<i>70.50 (0.00)</i>	91.23 (0.05)*
	ACC	65.07 (0.00)	61.30 (0.00)	66.42 (0.00)	<u>78.77 (0.00)</u>	40.51 (0.00)	42.10 (0.00)	<i>80.27 (0.00)</i>	94.13 (0.01)*
DSR	MAP	90.75 (0.00)	89.35 (0.00)	<u>92.54 (0.00)</u>	85.50 (0.00)	83.06 (0.00)	88.92 (0.00)	<i>93.00 (0.00)</i>	94.56 (0.02)*
	ACC	<u>96.73 (0.00)</u>	96.08 (0.00)	<i>97.39 (0.00)</i>	93.46 (0.00)	93.79 (0.00)	96.08 (0.00)	98.58 (0.00)	96.67 (0.06)*
GPA	MAP	99.98 (0.00)	86.90 (0.00)	67.39 (0.00)	99.06 (0.00)	99.53 (0.00)	99.06 (0.00)	<i>99.82 (0.00)</i>	<u>99.67 (0.01)†</u>
	ACC	99.37 (0.00)	90.51 (0.00)	56.01 (0.00)	<u>99.37 (0.00)</u>	<i>99.68 (0.00)</i>	<u>99.37 (0.00)</u>	100.00 (0.00)	100.00 (0.00)†
GPM	MAP	97.64 (0.00)	<u>98.58 (0.00)</u>	98.00 (0.00)	100.00 (0.00)	<i>99.53 (0.00)</i>	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)†
	ACC	98.42 (0.00)	99.05 (0.00)	98.00 (0.00)	100.00 (0.00)	<u>99.68 (0.00)</u>	100.00 (0.00)	100.00 (0.00)	<i>99.85 (0.03)†</i>
GPO	MAP	100.00 (0.00)	<i>96.31 (0.00)</i>	<u>45.05 (0.00)</u>	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)†
	ACC	100.00 (0.00)	<i>97.46 (0.00)</i>	<u>27.94 (0.00)</u>	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)†
Ham	MAP	69.99 (0.00)	69.18 (0.00)	69.02 (0.00)	69.15 (0.00)	69.07 (0.00)	<u>77.56 (0.00)</u>	<i>78.23 (0.00)</i>	81.22 (0.04)*
	ACC	68.57 (0.00)	69.52 (0.00)	<u>73.33 (0.00)</u>	71.43 (0.00)	66.67 (0.00)	<u>73.33 (0.00)</u>	<i>75.00 (0.00)</i>	80.95 (0.01)*
IERT	MAP	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)†
	ACC	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)†
IPD	MAP	<u>93.39 (0.00)</u>	92.20 (0.00)	80.12 (0.00)	93.27 (0.00)	90.62 (0.00)	92.50 (0.00)	<i>93.45 (0.00)</i>	96.33 (0.00)*
	ACC	95.04 (0.00)	95.63 (0.00)	83.58 (0.00)	95.72 (0.00)	95.52 (0.00)	<i>96.00 (0.00)</i>	<u>95.87 (0.00)</u>	96.21 (0.00)†
Meat	MAP	90.00 (0.00)	53.51 (0.00)	<i>94.28 (0.00)</i>	72.38 (0.00)	62.47 (0.00)	87.53 (0.00)	<u>91.50 (0.00)</u>	95.05 (0.04)†
	ACC	93.33 (0.00)	58.33 (0.00)	<u>91.67 (0.00)</u>	71.67 (0.00)	71.67 (0.00)	45.00 (0.00)	<i>92.00 (0.00)</i>	93.33 (0.00)†
MP	MAP	<u>89.47 (0.00)</u>	85.00 (0.00)	81.46 (0.00)	83.38 (0.00)	86.68 (0.00)	85.10 (0.00)	<i>90.00 (0.00)</i>	93.67 (0.02)*
	ACC	87.95 (0.00)	88.42 (0.00)	84.92 (0.00)	81.06 (0.00)	<u>92.62 (0.00)</u>	80.03 (0.00)	<i>93.00 (0.00)</i>	94.71 (0.02)*
MS2T	MAP	65.95 (0.00)	82.46 (0.00)	81.62 (0.00)	82.00 (0.00)	<u>86.13 (0.00)</u>	85.98 (0.00)	<i>88.33 (0.00)</i>	90.20 (0.04)*
	ACC	77.98 (0.00)	85.37 (0.00)	83.51 (0.00)	84.28 (0.00)	<u>91.38 (0.00)</u>	90.33 (0.00)	<i>92.05 (0.00)</i>	93.67 (0.05)*
MS	MAP	80.00 (0.00)	85.38 (0.00)	72.50 (0.00)	82.46 (0.00)	86.48 (0.00)	<u>86.50 (0.00)</u>	<i>89.11 (0.00)</i>	91.50 (0.05)*
	ACC	83.47 (0.00)	88.43 (0.00)	79.71 (0.00)	85.62 (0.00)	89.26 (0.00)	<u>89.67 (0.00)</u>	92.05 (0.00)	<i>89.69 (0.00)*</i>
O2	MAP	83.21 (0.00)	<u>88.00 (0.00)</u>	54.33 (0.00)	64.15 (0.00)	87.31 (0.00)	86.95 (0.00)	<i>88.42 (0.00)</i>	90.55 (0.06)*
	ACC	<i>86.67 (0.00)</i>	90.00 (0.00)	60.00 (0.00)	<u>66.67 (0.00)</u>	90.00 (0.00)	90.00 (0.00)	90.00 (0.00)	90.00 (0.02)†
Plane	MAP	<i>97.63 (0.00)</i>	93.25 (0.00)	95.56 (0.00)	81.34 (0.00)	95.05 (0.00)	95.00 (0.00)	97.94 (0.00)	<u>97.58 (0.04)†</u>
	ACC	100.00 (0.00)	<u>96.19 (0.00)</u>	100.00 (0.00)	84.76 (0.00)	100.00 (0.00)	<i>99.00 (0.00)</i>	100.00 (0.00)	<u>96.19 (0.00)*</u>
S2	MAP	94.01 (0.00)	98.46 (0.00)	98.90 (0.00)	96.61 (0.00)	<i>99.73 (0.00)</i>	99.46 (0.00)	<u>99.50 (0.00)</u>	100.00 (0.00)†
	ACC	89.33 (0.00)	94.67 (0.00)	<i>99.33 (0.00)</i>	92.00 (0.00)	<i>99.33 (0.00)</i>	<u>98.00 (0.00)</u>	<i>99.33 (0.00)</i>	100.00 (0.00)†
WEI	MAP	77.82 (0.00)	<u>84.46 (0.00)</u>	52.31 (0.00)	82.93 (0.00)	64.27 (0.00)	83.96 (0.00)	<i>86.91 (0.00)</i>	89.33 (0.01)*
	ACC	87.50 (0.00)	<u>96.85 (0.00)</u>	70.00 (0.00)	90.00 (0.00)	72.50 (0.00)	96.67 (0.00)	<i>99.00 (0.00)</i>	99.50 (0.02)†
MSA	MAP	<i>59.80 (0.00)</i>	58.50 (0.00)	57.32 (0.00)	52.95 (0.00)	44.25 (0.00)	58.33 (0.00)	<u>59.67 (0.00)</u>	63.50 (0.06)*
	ACC	82.97 (0.00)	<u>84.18 (0.00)</u>	80.32 (0.00)	81.82 (0.00)	73.67 (0.00)	84.00 (0.00)	<i>87.23 (0.00)</i>	88.00 (0.04)†
SAD	MAP	56.43 (0.00)	<u>62.71 (0.00)</u>	49.76 (0.00)	51.22 (0.00)	47.18 (0.00)	62.67 (0.00)	<i>64.93 (0.00)</i>	67.15 (0.08)*
	ACC	97.76 (0.00)	<u>97.89 (0.00)</u>	94.89 (0.00)	90.00 (0.00)	75.00 (0.00)	97.67 (0.00)	<i>98.71 (0.00)</i>	99.00 (0.00)†

BasicMotions (BM) dataset the performance of OTSW is not significantly different from that of the best-performing competitor, even when OTSW does not attain the highest scores.

Following several prior studies [18], [19], [88], we also construct a critical difference (CD) diagram to visualize the average rank of each distance measure across all datasets. This diagram additionally indicates whether the differences in performance between any pair of measures are statistically significant, based on the Wilcoxon signed-rank test with a significance level of $\alpha = 0.05$ [87]. Figure 4 presents two CD diagrams derived from the performance of

k -NN classifiers equipped with different distance measures, evaluated using MAP and ACC, respectively. The results show that the proposed distance consistently outperforms most baselines, particularly in terms of MAP, as illustrated in Figure 4(a).

These results validate the advantages of the proposed distance. The tree-sampling procedure in OTSW encodes both order differences and spatial distances between sequence elements into the structure of ordered trees. Consequently, the resulting tree metrics, which serve as the ground distances in OTSW, enable a more accurate measurement of distances between sequences. Existing OT-based distance measures

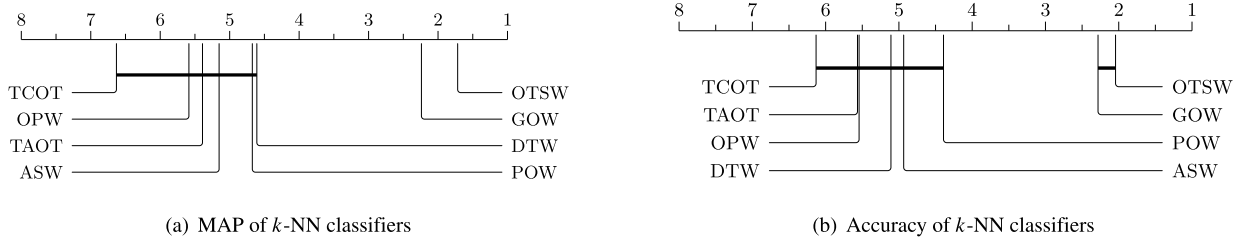


FIGURE 4. Critical difference diagrams showing the mean ranks, obtained from the Wilcoxon signed-rank test, of k -NN with different distance measures. The lower the rank (further to the right), the better the distance measure compared to the others, on average. Horizontal bold lines indicate that there is no significant difference in performance among the distance measures crossed by that particular line, in terms of the Friedman test that compares the ranks of multiple distance measures [87].

also attempt to incorporate order information, either by directly integrating it into the ground distance (e.g., TCOT and ASW) or by introducing regularization terms on the transport matrix (e.g., OPW, TAOT, POW, and GOW). However, such strategies modify the ground distance in ways that violate metric axioms, preventing them from being valid distance for sequences. As a result, their performance is generally inferior to that of OTSW. Our proposed distance also outperforms DTW, a widely used distance measure for sequences, in terms of both evaluation metrics. Because DTW imposes strict boundary and monotonicity constraints on sequence alignment and does not satisfy the triangle inequality, its contribution to improvement of the classification accuracy is limited.

We next empirically evaluate the computational efficiency of the proposed distance. We implemented k -NN with OTSW and baselines in Python with our best effort. We then executed them on a PC equipped with an Intel Xeon W CPU operating at 3.2 GHz. For each dataset, the execution was repeated five times. Table 3 reports the average computational time of the 1-NN classifier with different distance measures. Note that the last column of the table reports the running time of OTSW, which already includes the cost for sampling ordered trees. We can observe that OTSW is faster than all baselines in 17 out of 23 datasets. In particular, on large and long datasets such as CET and MS2T, it reduces runtime by an order of magnitude compared to most compared distance measures. Although DTW and TAOT are occasionally faster on a few datasets (e.g., MP, DSR, MS, O2), their overall average runtime remains substantially higher than OTSW. These observations further support our claims regarding the efficiency of the proposed distance.

Because OTSW leverages a closed-form solution to the optimal transport problem under tree metrics, its time complexity is linear w.r.t sequence length. In contrast, existing OT-based distance measures rely on Sinkhorn's matrix scaling algorithm, whose time complexity is quadratic in sequence length. In addition, Sinkhorn's algorithm is prone to producing very large scaling values that approach the numerical precision limits of the machine [17], [18], [19], which can slow down convergence in practice. Therefore, OTSW outperforms all existing OT-based distance measures

by a large margin in terms of computational efficiency. DTW remains a competitive baseline and is slightly faster than OTSW on three datasets. However, DTW employs a dynamic programming algorithm with quadratic time complexity w.r.t sequence length for sequence alignment. As a result, it is significantly slower than OTSW on most of the remaining datasets.

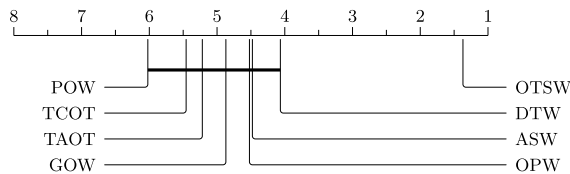
Clustering. We conduct a second set of experiments to evaluate the performance of OTSW in the time-series clustering task. The k -medoids algorithm is employed and combined with the proposed distance and baselines to partition time series from 23 datasets. Clustering performance is evaluated using NMI and ACC, and the results are reported in Table 4. We can observe that OTSW outperforms all existing distance measures on 14 out of the 23 datasets, regardless of the evaluation metric. For an additional 6 datasets, OTSW surpasses all baselines in terms of either NMI or ACC.

To further strengthen the evaluation, we again perform McNemar's test [86] to compare OTSW with the best-performing competitor on each dataset. The results are also included in Table 4. We can observe that on most datasets, where OTSW achieve the highest scores in terms of both evaluation metrics, it significantly improves clustering performance of k -medoids compared to baselines. On the remaining datasets, excepts CET and MSA, performance of OTSW is not significantly different from those of best-performing baselines, even when OTSW do not obtain the highest scores.

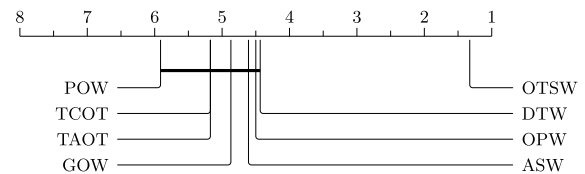
To provide a more comprehensive evaluation, we further rank the proposed distance and the baselines by constructing CD diagrams using the Wilcoxon signed-rank test with a significance level of $\alpha = 0.05$. The diagrams are shown in Figure 5. The results show that OTSW significantly outperform all existing distance measures, regardless of the evaluation metric. Unlike the classification setting, where labeled training data are available to support prediction, clustering operates without supervised information. As a result, the performance of the k -medoids algorithm depends strongly on how effectively the chosen distance measure captures the underlying cluster structure. However, many existing distance measures are not true metrics, as they violate the triangle inequality, thereby reducing their effectiveness

TABLE 3. Average computational time (in seconds) of 1-NN equipped with different distance measures over five times of execution. For each dataset, the fastest running time is highlighted in bold. The last row shows average computational times across over all datasets.

Datasets	DTW	OPW	TCOT	TAOT	ASW	POW	GOW	OTSW
AH	20.25	919.11	202.30	26.23	31.43	59.08	77.54	17.92
BM	10.03	33.05	23.36	6.94	5.50	7.92	14.64	4.40
BF	14.31	36.62	42.83	15.36	19.26	26.85	45.86	9.70
CBF	36.21	1976.09	531.99	72.39	54.84	165.61	194.24	27.72
CT	9.67	194.38	11.16	25.05	7.19	11.92	23.17	6.88
CET	2040.95	7809.40	8180.78	5976.84	8622.25	17085.20	25525.16	278.26
DSR	31.18	798.76	242.40	17.52	34.22	75.16	192.86	25.85
GPA	78.49	757.80	335.40	196.66	120.53	491.85	587.14	72.58
GPM	79.91	760.68	334.79	199.67	120.66	497.86	602.65	69.70
GPO	78.75	778.35	341.06	208.07	122.05	502.07	652.31	66.99
Ham	100.64	1490.13	2524.18	108.56	193.42	422.93	904.51	67.21
IERT	149.40	646.67	712.58	177.50	396.16	742.07	705.87	90.67
IPD	26.12	13935.93	1965.43	61.57	82.23	117.49	596.15	24.25
Meat	32.58	1018.87	830.30	30.69	50.85	126.07	273.86	30.92
MP	1248.00	7868.86	8244.41	8058.81	4209.37	4998.82	5898.21	1255.74
MS2T	475.46	3356.13	3659.34	2759.47	4167.37	6898.46	8942.35	263.42
MS	64.09	2384.65	440.53	28.48	52.63	73.91	105.48	47.59
O2	11.58	208.31	233.38	13.07	21.53	73.10	98.06	12.44
Plane	26.29	1525.22	401.06	23.44	32.31	84.00	98.26	16.89
S2	10.39	2785.06	71.40	25.41	40.75	45.56	63.82	2.25
WEI	55.03	161.69	139.88	124.77	263.59	159.60	198.04	56.92
MSA	67.91	132.21	168.55	148.97	215.65	159.32	199.23	26.51
SAD	754.47	444.19	466.02	443.34	437.65	502.63	647.58	35.07
Average	235.73	2174.88	1308.83	815.17	839.19	1449.02	2028.13	109.13



(a) NMI of k -medoids methods



(b) Accuracy of k -medoids methods

FIGURE 5. Critical difference diagrams showing the mean ranks, obtained from the Wilcoxon signed-rank test, of k -medoids with different distance measures. The lower the rank (further to the right), the better the distance measure compared to the others, on average. Horizontal bold lines indicate that there is no significant difference in performance among the distance measures crossed by that particular line, in terms of the Friedman test that compares the ranks of multiple distance measures [87].

in distinguishing between clusters. In contrast, the proposed distance is theoretically guaranteed to satisfy all metric properties. Consequently, it enables more reliable separation of time-series clusters and yields improved performance in clustering tasks.

We next empirically evaluate the computational efficiency of the proposed distance in the clustering task. Table 5 reports the average computational time of the k -medoids algorithm when combined with different distance measures across 23 datasets. The Python implementations of the proposed distance and the baselines are executed on each dataset five times using the same PC configuration as in the classification experiments. Note that the last column of the table presents the running time of OTSW, which already accounts for the cost of sampling ordered trees. The results show that OTSW is faster than all baseline methods on 21 out of the 23 datasets.

In particular, on large-scale datasets such as CET and SAD, OTSW accelerates the k -medoids algorithm by more than an order of magnitude compared with the best-performing baseline distance measures.

On average, existing OT-based distance measures are between 14 and 40 times slower than OTSW. Their primary limitation lies in their quadratic time complexity with respect to sequence length. Moreover, these measures rely on Sinkhorn's matrix scaling algorithm, which is prone to numerical instability due to machine-precision limits [17], [18], [19], thereby substantially slowing convergence in practice. In contrast, OTSW admits a closed-form solution to the optimal transport problem, resulting in significantly reduced computational time.

DTW is generally faster than existing OT-based distance measures, despite sharing the same quadratic time complexity

TABLE 4. Clustering results of the k -medoids algorithm with different distance measures. For each dataset, the algorithm was executed five times and the average scores along with standard deviation are reported. The top three highest scores for each dataset are presented in bold, italic red and underlined blue, respectively. The symbol † indicates that the performance of the proposed distance is not significantly different from the best performance from baselines at the $\alpha = 5\%$ level, using McNemar's test [86], while the symbol * indicates that it is significantly different.

Datasets		DTW	OPW	TCOT	TAOT	ASW	POW	GOW	OTSW
AH	NMI	<i>0.315 (0.000)</i>	0.178 (0.000)	0.180 (0.000)	<u>0.250 (0.000)</u>	0.219 (0.000)	0.190 (0.000)	0.200 (0.000)	0.521 (0.001)*
	ACC	<i>0.621 (0.000)</i>	0.592 (0.000)	0.592 (0.000)	<u>0.616 (0.000)</u>	0.578 (0.000)	0.588 (0.000)	0.600 (0.000)	0.775 (0.001)*
BM	NMI	0.811 (0.000)	0.585 (0.000)	0.542 (0.000)	0.192 (0.000)	<u>0.687 (0.000)</u>	0.307 (0.000)	0.331 (0.000)	<i>0.792 (0.012)†</i>
	ACC	0.887 (0.000)	0.738 (0.000)	0.675 (0.000)	0.425 (0.000)	<u>0.750 (0.000)</u>	0.512 (0.000)	0.587 (0.000)	<i>0.879 (0.007)†</i>
BF	NMI	0.030 (0.000)	<u>0.137 (0.000)</u>	<i>0.202 (0.000)</i>	0.047 (0.000)	0.010 (0.000)	0.019 (0.000)	0.021 (0.000)	0.283 (0.006)*
	ACC	0.600 (0.000)	<u>0.700 (0.000)</u>	<i>0.750 (0.000)</i>	0.625 (0.000)	0.500 (0.000)	0.575 (0.000)	0.602 (0.000)	0.830 (0.008)*
CBF	NMI	0.304 (0.000)	0.450 (0.000)	0.429 (0.000)	0.316 (0.000)	<i>0.647 (0.000)</i>	<u>0.561 (0.000)</u>	0.541 (0.000)	0.696 (0.001)*
	ACC	0.602 (0.000)	0.734 (0.000)	0.718 (0.000)	0.591 (0.000)	<i>0.841 (0.000)</i>	0.789 (0.000)	<u>0.801 (0.000)</u>	0.882 (0.000)*
CT	NMI	<i>0.160 (0.000)</i>	0.030 (0.000)	0.002 (0.000)	0.021 (0.000)	<u>0.035 (0.000)</u>	0.027 (0.000)	0.030 (0.000)	0.210 (0.000)*
	ACC	<i>0.758 (0.000)</i>	0.714 (0.000)	0.714 (0.000)	0.714 (0.000)	0.714 (0.000)	0.713 (0.000)	<u>0.715 (0.000)</u>	0.801 (0.000)*
CET	NMI	<i>0.232 (0.000)</i>	<u>0.088 (0.000)</u>	0.037 (0.000)	0.043 (0.000)	0.067 (0.000)	0.026 (0.000)	0.026 (0.000)	0.293 (0.000)*
	ACC	0.500 (0.000)	<u>0.400 (0.000)</u>	0.347 (0.000)	0.327 (0.000)	<u>0.400 (0.000)</u>	0.353 (0.000)	0.345 (0.000)	<i>0.422 (0.001)*</i>
DSR	NMI	0.806 (0.000)	0.866 (0.000)	0.768 (0.000)	<u>0.863 (0.000)</u>	0.770 (0.000)	0.785 (0.000)	0.805 (0.000)	<i>0.865 (0.006)†</i>
	ACC	0.894 (0.000)	<u>0.950 (0.000)</u>	0.857 (0.000)	<i>0.960 (0.000)</i>	0.882 (0.000)	0.885 (0.000)	0.853 (0.000)	0.973 (0.003)†
GPA	NMI	<i>0.090 (0.000)</i>	<u>0.070 (0.000)</u>	<u>0.070 (0.000)</u>	0.022 (0.000)	0.024 (0.000)	0.002 (0.000)	0.006 (0.000)	0.120 (0.024)*
	ACC	0.505 (0.000)	<i>0.645 (0.000)</i>	<i>0.645 (0.000)</i>	0.585 (0.000)	<u>0.590 (0.000)</u>	0.528 (0.000)	0.528 (0.000)	0.682 (0.012)*
GPM	NMI	0.341 (0.000)	0.536 (0.000)	0.536 (0.000)	0.062 (0.000)	0.675 (0.000)	0.438 (0.000)	<u>0.584 (0.000)</u>	<i>0.670 (0.012)†</i>
	ACC	0.743 (0.000)	<u>0.865 (0.000)</u>	<u>0.865 (0.000)</u>	0.648 (0.000)	0.922 (0.000)	0.814 (0.000)	0.833 (0.000)	<i>0.921 (0.004)†</i>
GPO	NMI	<i>0.344 (0.000)</i>	0.029 (0.000)	0.029 (0.000)	<u>0.217 (0.000)</u>	0.012 (0.000)	0.088 (0.000)	0.091 (0.000)	0.412 (0.003)*
	ACC	<u>0.745 (0.000)</u>	0.601 (0.000)	0.601 (0.000)	<i>0.767 (0.000)</i>	0.568 (0.000)	0.661 (0.000)	0.701 (0.000)	0.792 (0.003)*
Ham	NMI	0.022 (0.000)	0.030 (0.000)	<u>0.048 (0.000)</u>	0.010 (0.000)	0.001 (0.000)	0.043 (0.000)	<i>0.067 (0.000)</i>	0.195 (0.001)*
	ACC	0.584 (0.000)	0.603 (0.000)	<u>0.626 (0.000)</u>	0.556 (0.000)	0.519 (0.000)	0.622 (0.000)	<i>0.694 (0.000)</i>	0.755 (0.002)*
IERT	NMI	1.000 (0.000)	0.259 (0.000)	0.260 (0.000)	0.205 (0.000)	0.284 (0.000)	<u>0.359 (0.000)</u>	<i>0.495 (0.000)</i>	1.000 (0.006)†
	ACC	1.000 (0.000)	0.682 (0.000)	0.688 (0.000)	0.614 (0.000)	0.701 (0.000)	<u>0.749 (0.000)</u>	<i>0.839 (0.000)</i>	1.000 (0.002)†
IPD	NMI	0.020 (0.000)	0.035 (0.000)	0.021 (0.000)	<u>0.045 (0.000)</u>	<i>0.046 (0.000)</i>	0.012 (0.000)	0.019 (0.000)	0.088 (0.028)*
	ACC	0.524 (0.000)	<u>0.527 (0.000)</u>	0.524 (0.000)	0.524 (0.000)	<i>0.535 (0.000)</i>	0.507 (0.000)	0.512 (0.000)	0.553 (0.022)*
Meat	NMI	0.608 (0.000)	0.216 (0.000)	0.081 (0.000)	<u>0.313 (0.000)</u>	0.120 (0.000)	0.127 (0.000)	0.158 (0.000)	<i>0.605 (0.004)†</i>
	ACC	<i>0.817 (0.000)</i>	0.492 (0.000)	0.383 (0.000)	<u>0.617 (0.000)</u>	0.517 (0.000)	0.433 (0.000)	0.593 (0.000)	0.841 (0.006)*
MP	NMI	<i>0.574 (0.000)</i>	0.363 (0.000)	0.326 (0.000)	0.459 (0.000)	<u>0.463 (0.000)</u>	0.388 (0.000)	0.416 (0.000)	0.577 (0.003)†
	ACC	0.508 (0.000)	0.489 (0.000)	0.415 (0.000)	<i>0.587 (0.000)</i>	0.588 (0.000)	0.464 (0.000)	0.472 (0.000)	<u>0.585 (0.005)†</u>
MS2T	NMI	0.324 (0.000)	<i>0.451 (0.000)</i>	0.370 (0.000)	<u>0.419 (0.000)</u>	0.396 (0.000)	0.412 (0.000)	<i>0.451 (0.000)</i>	0.485 (0.001)*
	ACC	0.532 (0.000)	<i>0.623 (0.000)</i>	0.557 (0.000)	<u>0.580 (0.000)</u>	0.543 (0.000)	0.580 (0.000)	<u>0.595 (0.000)</u>	0.653 (0.002)*
MS	NMI	0.517 (0.000)	0.019 (0.000)	0.008 (0.000)	0.406 (0.000)	0.034 (0.000)	<u>0.533 (0.000)</u>	<i>0.541 (0.000)</i>	0.642 (0.001)*
	ACC	<u>0.895 (0.000)</u>	0.587 (0.000)	0.561 (0.000)	0.851 (0.000)	0.613 (0.000)	<i>0.909 (0.000)</i>	0.871 (0.000)	0.911 (0.001)†
O2	NMI	<i>0.524 (0.000)</i>	0.254 (0.000)	0.213 (0.000)	0.077 (0.000)	<u>0.361 (0.000)</u>	0.236 (0.000)	0.284 (0.000)	0.547 (0.006)*
	ACC	<i>0.750 (0.000)</i>	0.600 (0.000)	0.517 (0.000)	0.417 (0.000)	<u>0.650 (0.000)</u>	0.567 (0.000)	0.606 (0.000)	0.767 (0.004)*
Plane	NMI	0.779 (0.000)	<u>0.862 (0.000)</u>	<i>0.881 (0.000)</i>	0.789 (0.000)	0.745 (0.000)	0.857 (0.000)	<u>0.862 (0.000)</u>	0.901 (0.004)*
	ACC	0.700 (0.000)	0.852 (0.000)	0.852 (0.000)	0.691 (0.000)	0.571 (0.000)	<u>0.932 (0.000)</u>	0.958 (0.000)	<i>0.952 (0.007)†</i>
S2	NMI	0.380 (0.000)	<u>0.555 (0.000)</u>	<i>0.646 (0.000)</i>	0.530 (0.000)	0.547 (0.000)	0.450 (0.000)	0.492 (0.000)	0.683 (0.005)*
	ACC	0.680 (0.000)	0.813 (0.000)	<i>0.867 (0.000)</i>	0.833 (0.000)	<u>0.843 (0.000)</u>	0.680 (0.000)	0.715 (0.000)	0.897 (0.003)*
WEI	NMI	0.326 (0.000)	<u>0.526 (0.000)</u>	<u>0.526 (0.000)</u>	0.525 (0.000)	0.731 (0.000)	0.384 (0.000)	0.403 (0.000)	<i>0.729 (0.000)†</i>
	ACC	0.300 (0.000)	<u>0.488 (0.000)</u>	<u>0.488 (0.000)</u>	<u>0.488 (0.000)</u>	0.688 (0.000)	0.388 (0.000)	0.425 (0.000)	<i>0.668 (0.000)†</i>
MSA	NMI	<i>0.368 (0.000)</i>	0.309 (0.000)	0.302 (0.000)	0.274 (0.000)	0.400 (0.000)	0.107 (0.000)	0.169 (0.000)	<u>0.326 (0.000)*</u>
	ACC	<u>0.273 (0.000)</u>	0.236 (0.000)	0.231 (0.000)	0.212 (0.000)	<i>0.323 (0.000)</i>	0.102 (0.000)	0.102 (0.000)	0.352 (0.000)*
SAD	NMI	<u>0.296 (0.000)</u>	0.282 (0.000)	0.282 (0.000)	<i>0.466 (0.000)</i>	0.289 (0.000)	0.171 (0.000)	0.198 (0.000)	0.582 (0.000)*
	ACC	0.364 (0.000)	0.402 (0.000)	<u>0.403 (0.000)</u>	<i>0.503 (0.000)</i>	0.345 (0.000)	0.242 (0.000)	0.269 (0.000)	0.601 (0.000)*

with respect to sequence length. This is likely because DTW employs a relatively straightforward dynamic programming algorithm, rather than Sinkhorn's iterative matrix scaling procedure, which can converge slowly. Nevertheless, OTSW remains substantially faster than DTW, achieving an average speedup of up to 2 times, owing to its favorable linear time complexity.

Visualization. Our third set of experiments evaluates the performance of OTSW on the time-series visualization task. We employ t-SNE [73] and combines it with different distance measures to learn two-dimensional vectors for time series in each dataset. These vectors serve as representation

of time series in a new space that enable visualization. We use the SmoothSubspace (S2) dataset as an illustrative example and visualize the resulting representation vectors in Figure 6.

From the visualizations, it is evident that t-SNE combined with DTW, TAOT, and ASW fails to clearly separate the three classes in the dataset. OPW, TCOT, POW, and GOW perform better, as they enable t-SNE to distinguish class 3 from the remaining classes; however, the separation between class 1 and class 2 remains ambiguous. In contrast, OTSW outperforms all baseline methods by allowing t-SNE to map similar sequences close to one

TABLE 5. Average computational time (in seconds) of k -medoids with different distance measures over five times of execution. For each dataset, the fastest running time is highlighted in bold. The last row shows average computational times across over all datasets.

Datasets	DTW	OPW	TCOT	TAOT	ASW	POW	GOW	OTSW
AH	43.33	65.75	54.01	62.91	289.89	201.03	298.34	12.33
BM	11.02	3.58	4.32	5.74	38.53	11.77	19.28	1.72
BF	3.52	15.82	14.08	18.67	16.66	38.73	55.39	4.44
CBF	450.72	618.26	534.07	725.45	5259.96	2682.23	3591.37	46.26
CT	21.35	44.31	41.57	52.30	834.77	79.15	108.17	4.89
CET	1940.25	6562.00	6128.29	7203.99	9970.17	15196.88	18935.91	104.48
DSR	168.19	232.89	212.15	234.97	323.81	700.54	981.27	26.56
GPA	133.81	231.49	246.20	203.12	217.95	846.07	1021.34	19.04
GPM	135.26	221.78	245.78	203.94	215.21	839.78	1432.95	19.05
GPO	134.83	220.87	251.71	199.66	217.43	851.85	986.15	19.04
Ham	104.77	178.28	178.29	187.01	390.62	714.07	905.22	20.99
IERT	300.54	744.36	568.17	1501.03	2002.25	3042.11	3851.62	42.22
IPD	203.56	442.55	432.66	551.49	681.95	933.15	1093.62	31.66
Meat	33.39	48.93	40.15	55.34	64.37	153.16	201.42	11.64
MP	2280.95	4996.04	5711.64	6407.41	8265.25	15372.15	9036.14	316.67
MS2T	464.36	3896.51	4421.90	3369.08	4450.36	6744.64	8734.19	67.80
MS	653.70	899.70	1113.29	914.42	1629.10	2029.66	2696.17	56.61
O2	11.45	17.75	15.41	23.64	68.01	68.97	72.33	6.96
Plane	28.53	31.12	26.23	35.51	45.65	115.00	146.32	7.60
S2	13.35	34.07	40.78	42.48	51.73	100.11	12.47	3.17
WEI	53.96	43.62	244.91	239.76	480.75	403.92	515.67	67.39
MSA	96.32	224.66	162.81	188.84	250.08	266.26	301.18	62.46
SAD	913.61	432.30	238.92	251.45	272.63	416.51	488.27	86.30
Average	356.56	878.55	909.88	986.01	1566.83	2252.51	2412.38	45.18

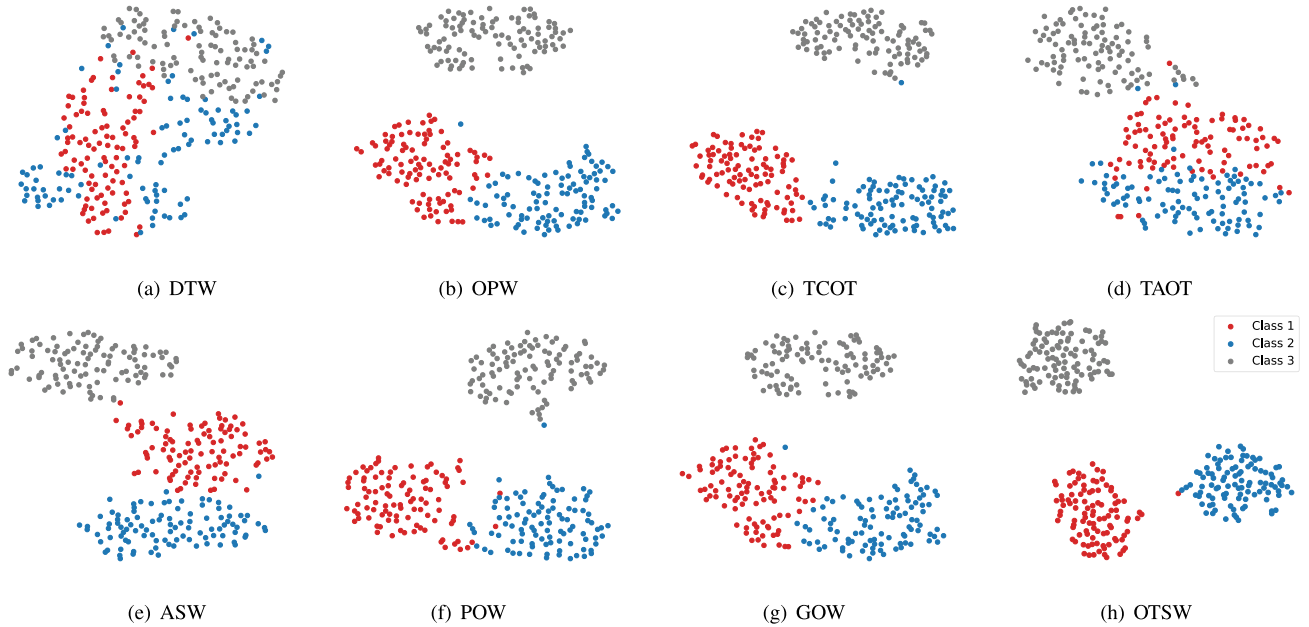


FIGURE 6. t-SNE visualization of sequences in SmoothSubspace dataset selected from UCR archive, using different distance measures. Different colors represent different classes.

another while clearly separating different classes with large margins.

To quantitatively evaluate the performance of t-SNE equipped with different distance measures, we adopt three

evaluation metrics: trustworthiness $T(5)$, continuity $C(5)$, and the local continuity meta-criterion $LCMC(5)$. The results across the 23 datasets are reported in Table 6. We observe that OTSW outperforms all baseline methods on 13 out of

TABLE 6. Quantitative results of the t-SNE with different distance measures. For each dataset, the method were executed five times and the average scores along with standard deviation are reported. The top three highest scores for each dataset are presented in bold, italic red and underlined blue, respectively. The symbol † indicates that the performance of the proposed distance is not significantly different from the best performance from baselines at the $\alpha = 5\%$ level, using McNemar's test [86], while the symbol * indicates that it is significantly different.

Datasets		DTW	OPW	TCOT	TAOT	ASW	POW	GOW	OTSW
AH	$T(5)$	97.22 (0.24)	94.79 (0.15)	95.24 (0.10)	94.46 (0.18)	95.58 (0.43)	95.57 (0.20)	<u>96.55 (0.14)</u>	<u>97.02 (0.31)†</u>
	$C(5)$	97.64 (0.11)	95.87 (0.09)	96.20 (0.09)	95.65 (0.08)	94.91 (0.32)	96.38 (0.05)	<u>96.42 (0.07)</u>	<u>97.04 (0.18)†</u>
	$LCMC(5)$	<u>61.79 (0.72)</u>	44.18 (0.54)	48.29 (0.51)	46.02 (0.13)	53.54 (0.90)	57.03 (0.67)	<u>57.94 (0.51)</u>	<u>61.98 (0.90)†</u>
BM	$T(5)$	92.05 (0.30)	97.52 (0.18)	<u>97.59 (0.15)</u>	97.02 (0.19)	<u>97.77 (0.12)</u>	96.47 (0.17)	96.82 (0.15)	<u>98.41 (0.08)†</u>
	$C(5)$	96.23 (0.17)	<u>97.93 (0.27)</u>	<u>97.90 (0.09)</u>	97.85 (0.19)	97.77 (0.10)	97.46 (0.12)	97.68 (0.10)	<u>98.71 (0.12)†</u>
	$LCMC(5)$	56.02 (1.60)	<u>58.22 (1.09)</u>	57.97 (1.07)	57.52 (1.01)	<u>58.57 (0.85)</u>	55.42 (0.94)	53.24 (0.09)	<u>64.42 (0.50)*</u>
BF	$T(5)$	<u>92.13 (0.32)</u>	91.66 (0.78)	91.03 (2.87)	<u>91.94 (0.75)</u>	93.68 (1.45)	90.90 (0.87)	91.00 (0.58)	89.09 (1.57)*
	$C(5)$	<u>90.87 (0.44)</u>	88.08 (1.96)	87.38 (3.18)	<u>90.81 (0.57)</u>	93.00 (2.23)	88.50 (0.91)	87.26 (0.87)	85.93 (1.38)*
	$LCMC(5)$	<u>55.38 (0.98)</u>	52.38 (1.89)	50.48 (3.89)	<u>55.98 (1.40)</u>	57.88 (2.14)	46.38 (1.21)	46.92 (0.99)	47.38 (2.11)*
CBF	$T(5)$	97.03 (0.03)	99.34 (0.00)	99.36 (0.01)	99.63 (0.00)	99.20 (0.02)	<u>99.50 (0.00)</u>	99.16 (0.01)	<u>99.38 (0.01)†</u>
	$C(5)$	97.09 (0.02)	99.20 (0.01)	<u>99.26 (0.01)</u>	99.55 (0.01)	99.10 (0.01)	<u>99.41 (0.01)</u>	99.11 (0.01)	99.22 (0.01)†
	$LCMC(5)$	30.17 (0.24)	51.02 (0.19)	<u>51.91 (0.26)</u>	53.83 (0.05)	49.15 (0.23)	50.56 (0.17)	51.19 (0.12)	<u>53.01 (0.17)†</u>
CT	$T(5)$	98.18 (0.03)	98.01 (0.08)	98.06 (0.12)	98.00 (0.16)	96.48 (0.19)	<u>98.39 (0.02)</u>	<u>98.32 (0.09)</u>	<u>98.41 (0.08)†</u>
	$C(5)$	97.05 (0.01)	97.13 (0.12)	97.11 (0.15)	97.10 (0.10)	94.99 (0.22)	97.96 (0.01)	<u>97.21 (0.24)</u>	<u>97.51 (0.13)†</u>
	$LCMC(5)$	53.43 (0.14)	49.77 (0.29)	49.96 (0.35)	49.24 (0.58)	45.54 (0.55)	<u>53.95 (0.23)</u>	<u>54.01 (0.19)</u>	<u>55.97 (0.44)*</u>
CET	$T(5)$	<u>99.69 (0.01)</u>	97.88 (0.03)	77.13 (0.68)	<u>98.99 (0.17)</u>	83.50 (0.67)	88.61 (0.34)	89.01 (0.45)	<u>99.73 (0.06)†</u>
	$C(5)$	<u>98.51 (0.03)</u>	95.09 (0.17)	78.20 (0.20)	<u>97.88 (0.07)</u>	85.59 (0.48)	84.80 (0.14)	85.88 (0.12)	<u>98.54 (0.09)†</u>
	$LCMC(5)$	<u>70.80 (0.10)</u>	52.74 (0.24)	4.89 (0.35)	<u>68.66 (0.41)</u>	16.73 (0.46)	17.98 (0.18)	25.67 (0.21)	<u>71.34 (0.53)*</u>
DSR	$T(5)$	<u>98.66 (0.04)</u>	93.76 (0.08)	95.27 (0.14)	94.68 (0.06)	96.50 (0.07)	<u>97.72 (0.05)</u>	97.54 (0.08)	<u>98.86 (0.06)†</u>
	$C(5)$	<u>98.48 (0.05)</u>	96.23 (0.07)	96.65 (0.04)	96.12 (0.07)	97.24 (0.08)	97.79 (0.08)	<u>98.03 (0.09)</u>	<u>98.65 (0.02)†</u>
	$LCMC(5)$	<u>61.72 (0.43)</u>	24.45 (0.44)	34.04 (0.31)	31.11 (0.33)	37.06 (0.45)	49.62 (0.33)	<u>50.05 (0.25)</u>	<u>62.43 (0.40)*</u>
GPA	$T(5)$	99.60 (0.02)	99.55 (0.01)	99.55 (0.01)	99.55 (0.00)	<u>99.68 (0.00)</u>	99.65 (0.00)	<u>99.67 (0.02)</u>	<u>99.74 (0.01)†</u>
	$C(5)$	99.48 (0.02)	99.54 (0.01)	99.54 (0.01)	99.53 (0.01)	<u>99.62 (0.01)</u>	<u>99.59 (0.02)</u>	99.45 (0.01)	<u>99.66 (0.01)†</u>
	$LCMC(5)$	<u>69.64 (0.09)</u>	63.15 (0.14)	63.07 (0.14)	62.69 (0.12)	<u>69.67 (0.31)</u>	66.48 (0.11)	67.15 (0.19)	<u>75.08 (0.16)*</u>
GPM	$T(5)$	99.61 (0.01)	99.53 (0.01)	99.53 (0.01)	99.53 (0.00)	<u>99.66 (0.01)</u>	<u>99.65 (0.00)</u>	99.51 (0.03)	<u>99.71 (0.01)†</u>
	$C(5)$	99.45 (0.05)	99.53 (0.02)	99.55 (0.00)	99.54 (0.00)	<u>99.61 (0.01)</u>	<u>99.56 (0.02)</u>	99.33 (0.02)	<u>99.71 (0.01)†</u>
	$LCMC(5)$	<u>69.44 (0.25)</u>	62.35 (0.07)	62.26 (0.13)	61.90 (0.11)	<u>69.60 (0.11)</u>	65.96 (0.15)	66.07 (0.11)	<u>74.64 (0.13)*</u>
GPO	$T(5)$	<u>99.60 (0.01)</u>	99.55 (0.01)	99.56 (0.00)	99.55 (0.01)	99.67 (0.00)	<u>99.66 (0.01)</u>	99.50 (0.02)	<u>99.66 (0.01)†</u>
	$C(5)$	99.41 (0.09)	<u>99.54 (0.00)</u>	<u>99.54 (0.01)</u>	99.52 (0.02)	99.61 (0.01)	<u>99.57 (0.02)</u>	99.50 (0.03)	<u>99.57 (0.02)†</u>
	$LCMC(5)$	<u>69.51 (0.23)</u>	62.92 (0.17)	63.04 (0.19)	62.53 (0.15)	<u>69.66 (0.28)</u>	66.03 (0.31)	66.15 (0.26)	<u>72.57 (0.29)*</u>
Ham	$T(5)$	95.07 (0.44)	<u>95.05 (0.28)</u>	94.22 (0.29)	91.21 (0.18)	92.86 (0.28)	94.20 (0.28)	94.25 (0.21)	<u>94.37 (0.17)†</u>
	$C(5)$	92.14 (0.51)	<u>93.15 (0.57)</u>	93.70 (0.20)	89.96 (0.59)	91.79 (0.78)	<u>92.63 (0.61)</u>	92.57 (0.52)	91.47 (0.44)*
	$LCMC(5)$	54.89 (0.62)	43.45 (0.64)	44.72 (0.62)	43.67 (0.67)	40.70 (0.93)	47.71 (0.71)	<u>48.05 (0.42)</u>	<u>49.15 (0.18)*</u>
IERT	$T(5)$	99.61 (0.01)	98.92 (0.05)	<u>99.42 (0.02)</u>	<u>99.34 (0.03)</u>	99.00 (0.03)	99.30 (0.01)	<u>99.42 (0.02)</u>	<u>99.61 (0.04)†</u>
	$C(5)$	99.39 (0.03)	99.11 (0.05)	99.19 (0.04)	99.20 (0.06)	99.18 (0.01)	<u>99.21 (0.02)</u>	99.15 (0.05)	<u>99.34 (0.06)†</u>
	$LCMC(5)$	<u>74.70 (0.14)</u>	60.35 (0.18)	<u>69.83 (0.49)</u>	68.78 (0.54)	69.26 (0.31)	<u>67.83 (0.33)</u>	68.52 (0.17)	<u>74.85 (0.29)†</u>
IPD	$T(5)$	98.37 (0.04)	97.34 (0.06)	97.98 (0.03)	97.76 (0.02)	<u>98.74 (0.01)</u>	<u>98.87 (0.02)</u>	98.52 (0.02)	<u>98.91 (0.03)†</u>
	$C(5)$	97.88 (0.02)	96.99 (0.01)	97.47 (0.01)	97.58 (0.01)	98.06 (0.01)	<u>98.46 (0.01)</u>	<u>98.51 (0.01)</u>	<u>98.97 (0.00)†</u>
	$LCMC(5)$	<u>49.99 (0.09)</u>	38.93 (0.19)	42.20 (0.16)	36.55 (0.04)	<u>49.00 (0.13)</u>	45.58 (0.18)	46.23 (0.24)	<u>52.16 (0.17)†</u>
Meat	$T(5)$	<u>98.54 (0.13)</u>	61.58 (0.32)	57.02 (0.18)	65.08 (0.32)	<u>75.91 (0.54)</u>	56.95 (1.16)	59.42 (1.08)	<u>98.93 (0.13)†</u>
	$C(5)$	<u>98.44 (0.23)</u>	83.10 (0.20)	75.97 (1.99)	<u>85.84 (0.31)</u>	83.85 (0.13)	78.48 (0.28)	79.50 (0.31)	<u>98.72 (0.07)†</u>
	$LCMC(5)$	<u>69.23 (0.83)</u>	13.86 (0.55)	12.60 (0.87)	20.16 (0.75)	23.80 (0.81)	25.10 (0.81)	<u>28.40 (0.52)</u>	<u>72.20 (0.57)*</u>
MP	$T(5)$	<u>99.37 (0.01)</u>	99.12 (0.03)	99.10 (0.01)	99.12 (0.02)	<u>99.59 (0.00)</u>	98.67 (0.02)	98.81 (0.01)	<u>99.80 (0.01)†</u>
	$C(5)$	<u>99.21 (0.00)</u>	93.10 (0.08)	93.09 (0.09)	93.16 (0.11)	<u>99.37 (0.00)</u>	92.78 (0.02)	93.88 (0.02)	<u>99.67 (0.01)†</u>
	$LCMC(5)$	59.84 (0.11)	27.53 (0.13)	27.58 (0.09)	27.57 (0.12)	<u>43.99 (0.05)</u>	27.74 (0.09)	28.61 (0.10)	<u>58.95 (0.06)†</u>
MS2T	$T(5)$	<u>98.37 (0.03)</u>	97.90 (0.10)	97.70 (0.12)	98.27 (0.06)	<u>98.31 (0.09)</u>	97.86 (0.13)	98.15 (0.09)	<u>98.81 (0.06)†</u>
	$C(5)$	97.28 (0.01)	96.14 (0.07)	96.14 (0.15)	<u>97.81 (0.03)</u>	<u>97.62 (0.04)</u>	97.05 (0.06)	97.15 (0.11)	<u>98.71 (0.26)*</u>
	$LCMC(5)$	55.06 (0.15)	58.01 (0.61)	58.70 (0.34)	<u>63.43 (0.42)</u>	<u>62.58 (0.24)</u>	62.06 (0.40)	62.37 (0.34)	<u>63.01 (0.13)†</u>
MS	$T(5)$	98.92 (0.06)	<u>99.51 (0.01)</u>	99.21 (0.06)	99.05 (0.04)	<u>99.33 (0.04)</u>	98.20 (0.08)	98.50 (0.07)	<u>99.57 (0.01)†</u>
	$C(5)$	<u>98.55 (0.02)</u>	<u>98.83 (0.01)</u>	97.67 (0.02)	98.47 (0.01)	96.82 (0.10)	94.81 (0.06)	95.01 (0.05)	<u>99.84 (0.02)†</u>
	$LCMC(5)$	<u>63.75 (0.16)</u>	62.50 (0.17)	55.52 (0.18)	<u>62.91 (0.19)</u>	57.23 (0.39)	48.67 (0.04)	49.12 (0.06)	<u>65.50 (0.11)*</u>
O2	$T(5)$	<u>96.61 (0.65)</u>	55.47 (0.82)	56.24 (2.02)	59.25 (1.53)	<u>74.78 (0.92)</u>	61.44 (2.04)	62.35 (1.12)	<u>97.45 (0.39)*</u>
	$C(5)$	<u>95.92 (0.55)</u>	74.24 (0.94)	71.24 (5.35)	79.86 (0.50)	<u>85.25 (0.41)</u>	79.91 (0.93)	80.17 (0.85)	<u>96.97 (0.26)*</u>
	$LCMC(5)$	<u>62.06 (1.87)</u>	11.26 (0.57)	1.99 (1.41)	14.73 (0.88)	<u>27.13 (0.71)</u>	19.73 (1.86)	21.33 (1.91)	<u>62.26 (1.10)†</u>
Plane	$T(5)$	<u>99.24 (0.03)</u>	98.75 (0.01)	99.03 (0.03)	98.77 (0.04)	99.03 (0.03)	<u>99.31 (0.05)</u>	99.15 (0.03)	<u>99.42 (0.06)†</u>
	$C(5)$	99.22 (0.01)	98.86 (0.03)	99.24 (0.03)	99.14 (0.01)	99.17 (0.05)	<u>99.46 (0.02)</u>	<u>99.45 (0.05)</u>	<u>99.83 (0.06)†</u>
	$LCMC(5)$	66.96 (0.30)	56.22 (0.23)	64.94 (0.97)	61.19 (0.81)	64.29 (0.46)	<u>70.33 (0.55)</u>	<u>71.05 (0.41)</u>	<u>72.45 (0.71)*</u>
S2	$T(5)$	95.63 (0.25)	<u>98.37 (0.06)</u>	<u>98.00 (0.04)</u>	94.33 (0.31)	96.95 (0.13)	97.10 (0.08)	97.93 (0.11)	<u>99.63 (0.17)*</u>
	$C(5)$	92.11 (0.85)	<u>97.18 (0.21)</u>	<u>96.87 (0.03)</u>	91.69 (0.25)	94.49 (0.21)	94.88 (0.10)	95.37 (0.18)	<u>97.83 (0.11)†</u>
	$LCMC(5)$	48.83 (0.86)	<u>55.74 (0.56)</u>	<u>53.97 (0.35)</u>	42.31 (0.38)	49.54 (0.50)	50.37 (0.49)	50.98 (0.33)	<u>59.82 (0.45)*</u>
WEI	$T(5)$	95.72 (0.44)	98.81 (0.08)	<u>98.83 (0.11)</u>	<u>98.93 (0.07)</u>	98.12 (0.12)	74.67 (0.51)	75.03 (0.31)	<u>99.43 (0.03)†</u>
	$C(5)$	97.22 (0.13)	<u>98.79 (0.06)</u>	<u>98.79 (0.05)</u>	<u>98.95 (0.05)</u>	98.49 (0.08)	83.54 (1.22)	85.67 (0.59)	<u>99.22 (0.04)†</u>
	$LCMC(5)$	66.62 (0.93)	75.02 (0.54)	<u>75.32 (0.72)</u>	<u>75.82 (0.98)</u>	72.67 (0.88)	31.27 (0.94)	35.12 (0.87)	<u>76.97 (0.70)*</u>
MSA	$T(5)$	98.60 (0.05)	98.88 (0.05)	<u>98.64 (0.02)</u>	97.75 (0.03)	98.08 (0.20)	58.76 (0.36)	59.84 (0.28)	<u>98.82 (0.04)†</u>
	$C(5)$	<u>99.13 (0.04)</u>	<u>99.07 (0.01)</u>	97.16 (0.10)	98.32 (0.02)	93.55 (0.07)	82.50 (0.22)	83.17 (0.17)	<u>99.34 (0.03)†</u>
	$LCMC(5)$	<u>71.38 (0.40)</u>	<u>73.08 (0.13)</u>	65.45 (0.20)	65.90 (0.16)	57.28 (0.29)	60.36 (0.38)	61.54 (0.21)	<u>77.72 (0.11)*</u>
SAD	$T(5)$	<u>99.18 (0.00)</u>	98.90 (0.04)	98.99 (0.04)	<u>99.02 (0.09)</u>	98.95 (0.01)	67.18 (0.60)	70.27 (0.86)	<u>99.54 (0.04)†</u>
	$C(5)$	<u>97.96 (0.02)</u>	97.83 (0.02)	97.84 (0.02)	<u>98.27 (0.11)</u>	98.57 (0.04)	81.91 (0.28)	83.29 (0.18)	<u>97.08 (0.03)*</u>
	$LCMC(5)$	56.24 (0.16)	56.19 (0.08)	<u>56.27 (0.07)</u>	<u>57.55 (0.16)</u>	55.60 (0.21)	56.24 (0.32)	55.81 (0.25)	<u>60.64 (0.20)*</u>

23 datasets, regardless of the evaluation metric used. On an additional 7 datasets, OTSW achieves the top score in at least one evaluation metric.

To further assess statistical significance, we perform McNemar's test [86] to compare OTSW with the best-performing baseline on each dataset. The results are also presented in Table 6. We observe that, in most cases where OTSW achieves the highest score in terms of $LCMC(5)$, its performance is significantly better than that of the second-best existing distance measure. For the other two metrics, $T(5)$ and $C(5)$, although OTSW outperforms the baselines on many datasets, the performance differences are generally not statistically significant.

A similar pattern is reflected in the critical difference diagrams shown in Figure 7, which are constructed using the Wilcoxon signed-rank test at a significance level of $\alpha = 0.05$. For $T(5)$ and $C(5)$, Figures 7(a) and 7(b) indicate that OTSW achieves the best average rank among all measures, although the differences are not statistically significant. In contrast, for $LCMC(5)$, OTSW attains the top rank, and its performance is significantly better than that of all existing distance measures.

Similar to time-series clustering, visualization using t-SNE is an unsupervised task, and the quality of the learned representation vectors depends heavily on the effectiveness of the underlying distance measures. The above results further demonstrate the advantages of OTSW over the existing distance measures. In particular, OTSW is a valid metric for sequences, whereas the baselines generally do not satisfy the triangle inequality. This theoretical property contributes to its enhanced ability to faithfully capture structural relationships among time series. It is worth noting that all three evaluation metrics quantify the agreement of local neighborhood structures between the original space and the embedding space. However, $LCMC(5)$ is symmetric, whereas $T(5)$ and $C(5)$ are directional.³ Therefore, the superior performance of OTSW in terms of $LCMC(5)$ provides strong evidence that combining OTSW with t-SNE effectively preserves the intrinsic local structure of the datasets.

We next evaluate the computational efficiency of the proposed distance and the baseline measures in the visualization task. Our Python implementations are executed using the same PC configuration as in the classification and clustering experiments. Table 7 reports the average running time of t-SNE combined with different distance measures on 23 time-series datasets. Note that the last column of the table contains computational time of OTSW, which already accounts for the cost of sampling ordered trees.

The results indicate that existing OT-based distance measures are, on average, slower than DTW, even though both DTW and these OT-based approaches have quadratic time complexity w.r.t sequence length. The primary reason is that OT-based methods rely on Sinkhorn's algorithm

to compute the optimal transport matrix. This iterative matrix-scaling procedure is prone to slow convergence, as it may generate extremely large scaling values that exceed machine-precision limits and must be numerically truncated, thereby increasing computational overhead. In contrast, DTW employs a straightforward dynamic programming algorithm to compute sequence alignments and distances, which reduces its practical running time. Indeed, DTW achieves the fastest running time on three datasets; however, these datasets are relatively small. For the remaining datasets, OTSW proves to be more efficient, consistently accelerating t-SNE compared with all baseline measures. On small datasets, the additional time required to sample ordered trees in OTSW may slightly increase its running time relative to DTW. However, as dataset size grows, the linear time complexity of OTSW for computing pairwise distances becomes increasingly advantageous. When distances must be computed for many pairs of sequences, as required by t-SNE, this linear complexity substantially reduces the overall computational time.

F. ABLATION STUDY

We finally study the influence of hyperparameters on the performance of OTSW. The proposed distance involves four key hyperparameters: τ (see Equation (5)), which balances the spatial and positional distances between two sequence elements during tree sampling; κ , which is the number of clusters in tree-sampling procedure (see Algorithm 2); $H_{\mathcal{T}}$, which specifies the maximum depth of the ordered trees (see Algorithm 1); and t , which is the number of ordered trees used to compute OTSW distance (see Equation (7)). To analyze their effects, we vary the value of one parameter while fixing the others and record the performance of the 1-NN classifier equipped with OTSW on the ItalyPowerDemand dataset selected from the UCR archive.

Influence of τ . Figure 8(a) and 8(b) depicts the classification results in terms of MAP and ACC averaged over five runs. In these experiments, τ varies from 10^{-3} to 10^4 , while κ is fixed at 2, and both $H_{\mathcal{T}}$ and t are set to 15. As shown in these figures, OTSW performs poorly when $\tau \leq 0.1$, likely due to the insufficient incorporation of order information during the tree-sampling process. The performance of OTSW improves sharply as τ increases beyond this value and reaches its peak at $\tau = 10$, highlighting the importance of incorporating the order information from sequence elements into the tree-sampling procedure. However, as τ continues to increase, the classification performance declines and drops significantly when $\tau > 10^2$. When τ becomes excessively large, the term of difference in order dominates the spatial dissimilarity in Equation (5). Consequently, the sampled trees may fail to adequately capture spatial relationships among elements, leading to degraded classification performance. We also report the running time required to sample a single ordered tree as τ varies in Figure 8(c). We can observe that as τ increases, the computational cost of tree sampling decreases. This behavior is likely because the increasing contribution

³ $C(5)$ measures the preservation of local structures from the original space to the embedding space, while $T(5)$ assesses preservation in the reverse direction.

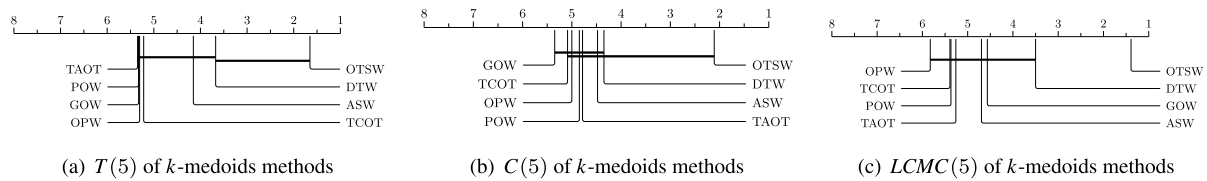


FIGURE 7. Critical difference diagrams showing the mean ranks, obtained from the Wilcoxon signed-rank test, of t-SNE with different distance measures. The lower the rank (further to the right), the better the distance measure compared to the others, on average. Horizontal bold lines indicate that there is no significant difference in performance among the distances crossed by that particular line, in terms of the Friedman test that compares the ranks of multiple distances [87].

TABLE 7. Average computational time (in seconds) of t-SNE with different distance measures over five times of execution. For each dataset, the fastest running time is highlighted in bold. The last row shows average computational times over all datasets.

Datasets	DTW	OPW	TCOT	TAOT	ASW	POW	GOW	OTSW
AH	63.31	89.08	81.54	100.37	85.77	96.32	165.72	45.10
BM	13.28	10.22	7.09	8.13	10.64	12.12	24.38	6.05
BF	5.33	12.16	13.50	15.84	11.64	13.82	21.42	10.47
CBF	602.73	936.92	750.45	907.78	922.72	1029.60	1250.67	487.52
CT	32.70	175.04	155.46	182.76	171.70	190.04	210.60	19.96
CET	3896.72	9622.14	9578.02	9599.54	9644.20	9628.50	12087.35	324.18
DSR	193.47	314.90	279.56	318.62	309.84	342.30	398.72	116.68
GPA	160.10	403.04	366.25	398.76	394.90	434.80	491.27	106.86
GPM	161.39	402.04	357.96	392.86	393.66	430.92	499.15	107.31
GPO	170.62	402.86	368.12	395.92	394.60	432.62	502.37	109.88
Ham	129.27	256.56	252.91	287.62	252.08	278.82	328.17	73.10
IERT	362.86	745.48	731.60	795.42	737.70	812.06	906.42	202.24
IPD	275.83	578.80	672.58	662.06	568.58	627.34	717.08	194.01
Meat	39.18	57.44	54.03	62.84	56.58	63.16	74.51	37.18
MP	10510.22	14033.78	12127.56	13098.88	13940.36	15458.52	15942.13	5516.86
MS2T	4916.01	4212.54	4186.99	4730.74	4179.84	4644.16	4896.17	313.40
MS	515.75	1508.88	1798.10	1526.92	1500.84	1658.50	1946.13	426.68
O2	13.63	25.80	24.12	27.96	25.36	28.48	35.42	19.88
Plane	34.21	43.64	38.99	45.06	42.98	47.52	51.67	25.81
S2	21.63	71.88	59.17	68.04	70.94	78.26	80.16	11.52
WEI	52.21	108.84	191.36	190.35	408.40	189.38	221.15	67.70
MSA	96.93	149.05	207.52	205.93	291.43	214.40	283.46	36.74
SAD	1237.68	412.57	477.07	474.39	492.76	559.11	601.52	103.29
Average	1110.02	1516.07	1465.04	1547.60	1574.20	1704.55	1897.63	375.19

of the second term in Equation (5) makes the distances between sequence elements more uniform, thereby inducing more balanced trees during the tree-sampling procedure. As a result, the computational time for tree sampling decreases and converges to a stable value as τ increases.

Influence of κ . To analyze the influence of κ on the performance of OTSW, we fix τ at 10 and set both $H_{\mathcal{T}}$ and t to 15, while varying κ from 2^1 to 2^{12} . Figures 8(d) and 8(e) show the average MAP and ACC scores of the 1-NN classifier equipped with OTSW over five runs. The results indicate that classification performance remains relatively stable when $\kappa \leq 2^9$. However, as κ increases beyond this value, the classification performance declines rapidly. This behavior can be attributed to the fact that a large number of clusters fragments natural data groupings into smaller clusters, thereby disrupting local neighborhood structures and distorting the tree distances between sequence elements. The number of clusters also affects the

computational time of the tree-sampling procedure, as illustrated in Figures 8(f). The time required to sample a tree gradually increases as κ grows from 2 to 2^6 , which is expected because the farthest-point clustering algorithm requires more time to partition sequence elements into a larger number of clusters. However, as κ continues to increase, the computational time becomes unstable. This is likely because an excessively large number of clusters causes the sampled trees to become shallower and induces skewed tree structures.

Influence of $H_{\mathcal{T}}$. We next fix $\tau = 10$, $\kappa = 2$, and $t = 15$, and study performance of the 1-NN classifier equipped with OTSW as $H_{\mathcal{T}}$ varies from 5 to 30. The average classification scores in terms of MAP and ACC over five runs are depicted in Figures 8(g) and 8(h), respectively. The results indicate that, for $H_{\mathcal{T}}$ within the range [5, 15], classification performance improves as the maximum depth increases. This behavior is expected, as deeper ordered trees can better

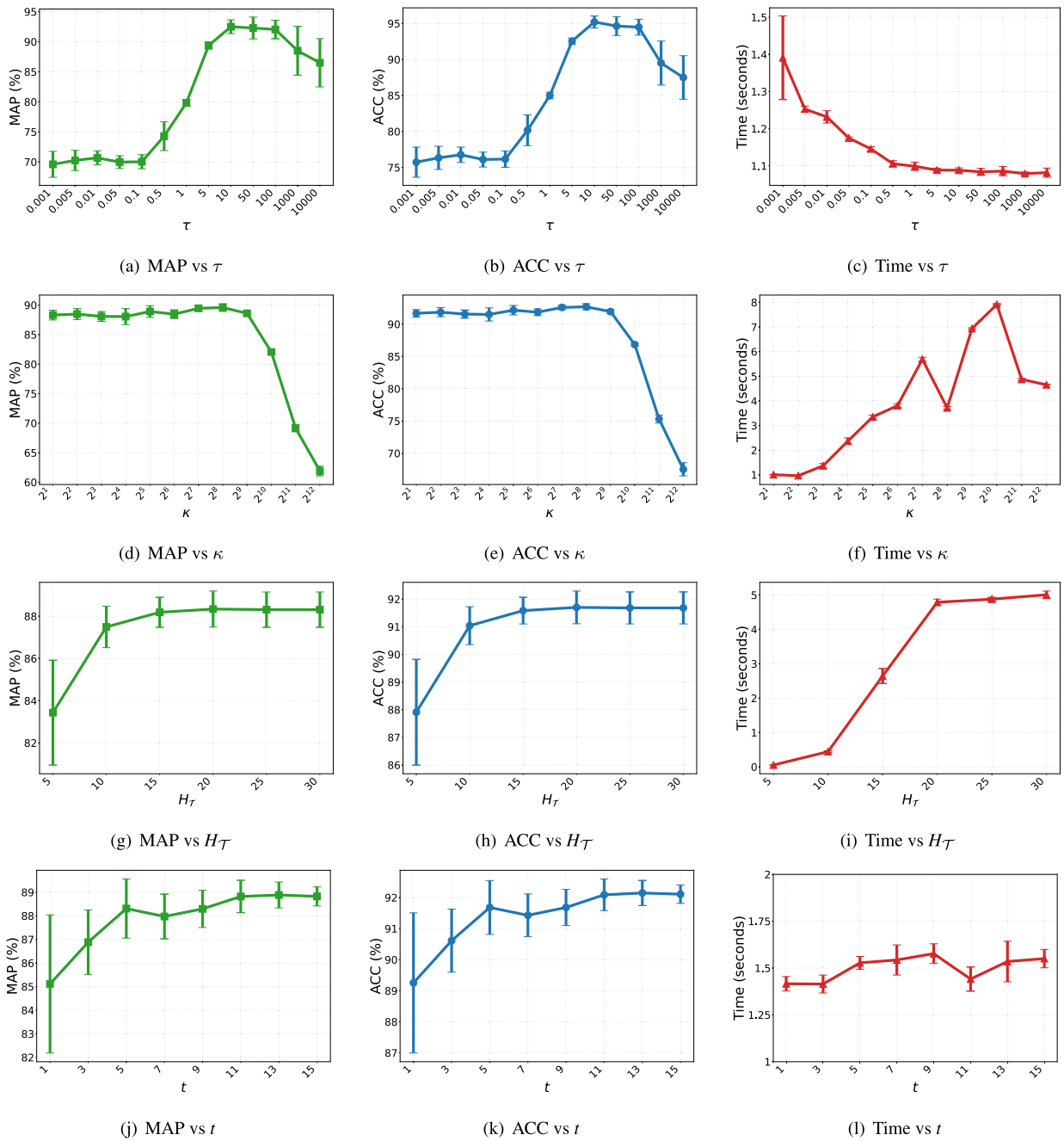


FIGURE 8. Ablation study results: performance of 1-NN classifier equipped with OTSW in terms of MAP and ACC, and time for sampling an ordered tree when hyperparameters, including τ , κ , H_T , and t , are varying.

represent the hierarchical relationships among sequence elements. However, when H_T exceeds 15, the classification performance stabilizes and shows no further improvement. We also examine the computational time required for tree sampling, as illustrated in Figure 8(i). The results show that as H_T increases, the time required to sample an ordered tree also increases. These observations suggest that H_T should be set to a moderate value to maintain strong performance while avoiding unnecessary computational overhead.

Influence of t . Finally, we vary t from 1 to 15, while fixing $\tau = 10$, $\kappa = 2$, and $H_T = 15$. The performance of OTSW in terms of MAP and ACC as t varies is illustrated in Figure 8(j) and 8(k) respectively. The results show that as the number of ordered trees increases from 1 to 5, the performance of OTSW improves rapidly. As t increases further, the performance stabilizes and shows no significant improvement. This behavior is expected, as the performance of OTSW with only a few ordered trees may be biased by

some low-quality samples. With a sufficient number of trees, the averaging operator in OTSW mitigates such bias, thereby enhancing the performance. In addition, the variance of the classification results decreases as t increases, indicating that the sampled trees contribute to stabilizing the performance of the proposed distance. Figure 8(1) shows the running time required to sample a single tree using Algorithm 1. As expected, t has no effect on the time required for sampling an ordered tree.

VII. CONCLUSION

In this paper, we propose a novel distance, called the *Ordered Tree-Sliced Wasserstein* (OTSW), for sequential data. OTSW is equipped with a tree-sampling procedure in which the order information of sequence elements is incorporated into the structure of each sampled tree by considering their relative positional differences. The sampled trees induce tree metrics, which serve as the ground distances for OTSW. By leveraging a closed-form solution to the optimal transport problem under tree metrics, OTSW achieves linear time complexity w.r.t sequence length. Further theoretical analysis reveals that OTSW is a true metric for sequences, as it satisfies all metric axioms. We extensively evaluate the applicability of OTSW in time-series classification, clustering, and visualization tasks. The experimental results validate its advantages over existing baselines.

Although OTSW demonstrates promising results, several directions for future research remain. First, the tree-sampling process has not yet been explored in the presence of supervised information. In supervised or semi-supervised settings, available labels could be leveraged to learn or sample more discriminative trees, potentially improving the performance of OTSW in downstream tasks. Second, the applicability of OTSW has not yet been examined across a broader range of tasks. For example, time-series prediction [89], anomaly detection [90], and segmentation [91] often require effective distance measures. Extending the expressiveness of OTSW could therefore be beneficial for such applications.

APPENDIX A OTBASED DISTANCE MEASURES FOR SEQUENTIAL DATA

In this section, we review formulations of existing OT-based distance measures for sequential data. Given two sequences $\mathcal{X}_u = \{\mathbf{x}_i^u\}_{i=1}^{n_u}$ and $\mathcal{X}_v = \{\mathbf{x}_i^v\}_{i=1}^{n_v}$, formulations of existing OT-based distance measures typically involve a ground distance matrix $\mathbf{D} \in \mathbb{R}^{n_u \times n_v}$, where each entry $d_{i,j}$ presents the difference between $\mathbf{x}_i^u$ and $\mathbf{x}_j^v$ measured by a ground metric (e.g. Euclidean distance), and a transport matrix $\mathbf{T} \in \mathbb{R}^{n_u \times n_v}$ that belongs to $\Pi(\mathbf{u}_{n_u}, \mathbf{u}_{n_v}) = \left\{ \mathbf{T} \in \mathbb{R}_+^{n_u \times n_v} \mid \mathbf{T} \mathbf{1}_{n_v} = \mathbf{u}_{n_u}, \mathbf{T}^\top \mathbf{1}_{n_u} = \mathbf{u}_{n_v} \right\}$. To incorporate order information of sequence elements into the formulation, existing OT-based distance measures either directly modify the ground distance matrix $\mathbf{D}$ [17], [23] or introduce regularizations on the transport matrix $\mathbf{T}$ [17], [18], [19], [24]. In general, these distance measures share the

following formulation:

$$\mathbf{T}^* = \underset{\mathbf{T} \in \Pi(\mathbf{u}_N, \mathbf{u}_M)}{\operatorname{argmin}} (\tilde{\mathbf{D}}, \mathbf{T})_F + \langle \mathbf{R}, \mathbf{T} \rangle_F - \frac{1}{\lambda} h(\mathbf{T}), \quad (19)$$

where $\tilde{\mathbf{D}}$ denotes the modified ground distance matrix, $\langle \mathbf{R}, \mathbf{T} \rangle_F$ presents the regularization term related to the order information, and $h(\mathbf{T})$ denotes entropic regularization term with the corresponding parameter λ .

For the distance measures that directly modifies the ground distance matrix, such as Temporally Coupled Optimal Transport (TCOT) [17] and Auto-weighted Sequential Wasserstein (ASW) distance [23], the matrix $\mathbf{R}$ is set to zero matrix, and the modified ground distance matrix $\tilde{\mathbf{D}}$ is constructed from $\mathbf{D}$.

Temporally Coupled Optimal Transport. Each entry $\tilde{d}_{i,j}$ of $\tilde{\mathbf{D}}$ in TCOT is computed as follows:

$$\tilde{d}_{i,j} = d_{i,j} \left(\left| \frac{i}{n_u} - \frac{j}{n_v} \right| + 1 \right). \quad (20)$$

Auto-weighted Sequential Wasserstein. Each entry $\tilde{d}_{i,j}$ of $\tilde{\mathbf{D}}$ in ASW is computed as follows:

$$\tilde{d}_{i,j} = d_{i,j} + w_1 \sum_{k=-l}^l \|\mathbf{x}_{i+k}^{u'} - \mathbf{x}_{j+k}^{v'}\|_2^2 + \frac{w_2}{1 + e^{-\alpha(|i-j|-\beta)}}, \quad (21)$$

where $\mathbf{x}_{i+k}^{u'}$ and $\mathbf{x}_{j+k}^{v'}$ denotes the second-order derivatives of the sequences at positions $i+k$ and $j+k$, respectively; l is a prespecified size of windows centered at $\mathbf{x}_i^u$ and $\mathbf{x}_j^v$; and α and β represent the steepness and midpoint of the sigmoid function, respectively. The weights w_1 and w_2 are automatically determined during optimization.

For the regularization-based distance measures, such as Order-Preserving Wasserstein (OPW) [17], Time-Adaptive Optimal Transport (TAOT) [24], Partial Ordered Wasserstein (POW) [18], and Generalized Ordered Wasserstein (GOW) [19], the ground distance matrix is remained unchanged ($\tilde{\mathbf{D}} = \mathbf{D}$). These methods differ primarily in how the matrix $\mathbf{R}$ is constructed.

Order Preserving Wasserstein. Each entry $r_{i,j}$ of the matrix $\mathbf{R}$ in OPW is computed as follows:

$$r_{i,j} = \frac{-\lambda_1^{-1}}{\left(\frac{i}{n_u} - \frac{j}{n_v}\right)^2 + 1} + \lambda_2 \left(\frac{l_{i,j}^2}{2\sigma^2} + \log(\sigma\sqrt{2\pi}) \right), \quad (22)$$

where $\lambda_1 > 0$ and $\lambda_2 > 0$ are additional regularization parameters, and $l_{i,j} = \frac{|i/n_u - j/n_v|}{\sqrt{1/n_u^2 + 1/n_v^2}}$.

Time-Adaptive Optimal Transport. Each entry $r_{i,j}$ of the matrix $\mathbf{R}$ in TAOT is computed as follows

$$r_{i,j} = (i - j)^2. \quad (23)$$

Partial Ordered Wasserstein. Each entry $r_{i,j}$ of the matrix $\mathbf{R}$ in POW⁴ is computed as follows:

$$r_{i,j} = \lambda_1 \left(\frac{i}{n_u} - \frac{j}{n_v} \right)^2. \quad (24)$$

Generalized Ordered Wasserstein. Each entry $r_{i,j}$ of the matrix $\mathbf{R}$ in GOW is computed as follows:

$$r_{i,j} = \lambda_1 \left(i - \sum_{k=1}^K w_k f_k(j) \right)^2, \quad (25)$$

where $f_k(\cdot)$ denotes a prespecified monotonic function and w_k represents its corresponding weight.

The formulation of the regularization-based distance measures can be rewritten as follows:

$$\begin{aligned} \mathbf{T}^* &= \underset{\mathbf{T} \in \Pi(\mathbf{u}_N, \mathbf{u}_M)}{\operatorname{argmin}} \langle \mathbf{D}, \mathbf{T} \rangle_F + \langle \mathbf{R}, \mathbf{T} \rangle_F - \frac{1}{\lambda} h(\mathbf{T}) \\ \Leftrightarrow \mathbf{T}^* &= \underset{\mathbf{T} \in \Pi(\mathbf{u}_N, \mathbf{u}_M)}{\operatorname{argmin}} \langle \mathbf{D} + \mathbf{R}, \mathbf{T} \rangle_F - \frac{1}{\lambda} h(\mathbf{T}), \end{aligned} \quad (26)$$

which is identical to the formulation of entropic regularized OT [26] with a new ground distance matrix $\mathbf{D} + \mathbf{R}$. This observation indicates that regularization-based distance measures implicitly alter the original ground distance matrix. Since all existing OT-based distance measures for sequences modify the metric properties of the ground distance, either explicitly or implicitly, they fail to satisfy the conditions of a true metric as stated in Proposition 2.3 of [22].

ACKNOWLEDGEMENT

This research is funded by Hanoi University of Science and Technology (HUST) under project number T2024-TĐ-025.

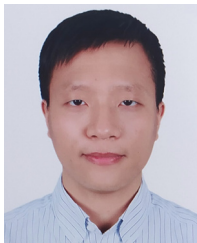
REFERENCES

- [1] G. Cheng, "A progressive fusion framework of rotationally-enhanced 3-D CNN and GCN-DTW for table tennis action recognition and quality evaluation," *IEEE Access*, vol. 14, pp. 24480–24495, 2026.
- [2] L. Li, M. Wang, and Y. Pi, "A random degradation aggregation network with temporal-spatial attention for satellite video super-resolution," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 19191–19207, 2025.
- [3] H. Fei, S. Wu, M. Zhang, M. Zhang, T.-S. Chua, and S. Yan, "Enhancing video-language representations with structural spatio-temporal alignment," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 7701–7719, Dec. 2024.
- [4] T. H. Vuong, T. Doan, and A. Takasu, "Deep wavelet convolutional neural networks for multimodal human activity recognition using wearable inertial sensors," *Sensors*, vol. 23, no. 24, p. 9721, Dec. 2023.
- [5] D. Ko, J. Choi, J. Ko, S. Noh, K.-W. On, E.-S. Kim, and H. J. Kim, "Video-text representation learning via differentiable weak temporal alignment," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5006–5015.
- [6] X. Chang, F. Tung, and G. Mori, "Learning discriminative prototypes with dynamic time warping," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8395–8404.
- [7] T. Wen Wang, Y. S. Lu, C. S. Ku, O. Mohammed, B. Raj, and L. Y. Por, "A multimodal deep fusion method for stock movement prediction using heterogeneous data source," *IEEE Access*, vol. 14, pp. 9882–9896, 2026.
- [8] B. Amiri, A. Haddadi, and K. Farajpour Mojdehi, "A novel hybrid GCN-LSTM algorithm for energy stock price prediction: Leveraging temporal dynamics and inter-stock relationships," *IEEE Access*, vol. 13, pp. 24815–24832, 2025.
- [9] M. R. A. Jamaludin, A. Lee, and C. S. Kim, "Enhanced quantum-based DNA sequence alignment with noise handling and error detection," *IEEE Access*, vol. 13, pp. 197371–197380, 2025.
- [10] O. Degachi, L. E. Amraoui, and K. Ouni, "ECG heartbeat classification using CNN autoencoder feature extraction and attention-augmented BiLSTM classifier," *IEEE Access*, vol. 13, pp. 171241–171254, 2025.
- [11] X. Huang, D. Wu, and B. Boulet, "Traffic signal control using lightweight transformers: An offline-to-online RL approach," *IEEE Open J. Intell. Transp. Syst.*, vol. 7, pp. 396–411, 2026.
- [12] N. Faqir, Y. Ennaji, L. Chakir, and J. Boumhidi, "Hybrid CNN-LSTM and proximal policy optimization model for traffic light control in a multi-agent environment," *IEEE Access*, vol. 13, pp. 29577–29588, 2025.
- [13] H. Wang, M. Saito, J. Liu, and S. Shimamoto, "Comprehensive analysis of traffic operation for on-street parking based on the urban road network," *IEEE Access*, vol. 12, pp. 125643–125653, 2024.
- [14] V. T. Hong, A. Takasu, and T. Doan, "Deep sensor-fusion approach to vehicle detection on bridges using multiple strain sensors," in *Life-Cycle of Structures and Infrastructure Systems*. Boca Raton, FL, USA: CRC Press, 2023, pp. 1935–1942.
- [15] V. T. Hong, T. Doan, and A. Takasu, "Sensor data alignment for multi-view bridge monitoring," in *Bridge Safety, Maintenance, Management, Life-Cycle, Resilience and Sustainability*. Boca Raton, FL, USA: CRC Press, 2022, pp. 1598–1606.
- [16] H. Sakoe and S. Chiba, "Dynamic programming algorithm optimization for spoken word recognition," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. ASSP-26, no. 1, pp. 43–49, Feb. 1978.
- [17] B. Su and G. Hua, "Order-preserving optimal transport for distances between sequences," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 12, pp. 2961–2974, Dec. 2019.
- [18] T. Doan, T. Phan, P. Nguyen, K. Than, M. Visani, and A. Takasu, "Partial ordered Wasserstein distance for sequential data," *Neurocomputing*, vol. 595, Aug. 2024, Art. no. 127908.
- [19] T. Doan, H. To, H. Vuong, M. Visani, A. Takasu, and K. Than, "Generalized ordered Wasserstein distance for sequential data," *Pattern Recognit.*, vol. 169, Jan. 2026, Art. no. 111952.
- [20] A. Abanda, U. Mori, and J. A. Lozano, "A review on distance based time series classification," *Data Mining Knowl. Discovery*, vol. 33, no. 2, pp. 378–412, Mar. 2019.
- [21] C. Holder, M. Middlehurst, and A. Bagnall, "A review and evaluation of elastic distance functions for time series clustering," *Knowl. Inf. Syst.*, vol. 66, no. 2, pp. 765–809, Feb. 2024.
- [22] P. Gabriel and M. Cuturi, *Computational Optimal Transport: With Applications to Data Science*. Boston, MA, USA: Now, 2019.
- [23] M. Horie and H. Kasai, "Auto-weighted sequential Wasserstein distance and application to sequence matching," in *Proc. 30th Eur. Signal Process. Conf. (EUSIPCO)*, Aug. 2022, pp. 1472–1476.
- [24] Z. Zhang, P. Tang, and T. Corpetti, "Time adaptive optimal transport: A framework of time series similarity measure," *IEEE Access*, vol. 8, pp. 149764–149774, 2020.
- [25] R. Sinkhorn and P. Knopp, "Concerning nonnegative matrices and doubly stochastic matrices," *Pacific J. Math.*, vol. 21, no. 2, pp. 343–348, May 1967.
- [26] C. Marco, "Sinkhorn distances: Lightspeed computation of optimal transport," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 26, 2013, pp. 2292–2300.
- [27] E. Vidal Ruiz, "An algorithm for finding nearest neighbours in (approximately) constant average time," *Pattern Recognit. Lett.*, vol. 4, no. 3, pp. 145–157, Jul. 1986.
- [28] H. Shimodaira, K.-I. Noma, M. Nakai, and S. Sagayama, "Dynamic time-alignment kernel in support vector machine," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 14, 2001, pp. 921–928.
- [29] H. Narita, Y. Sawamura, and A. Hayashi, "Learning a kernel matrix for time series data from DTW distances," in *Proc. Adv. Neural Inf. Process. Syst.*, 2007, pp. 336–345.
- [30] S. Salvador and P. Chan, "Toward accurate dynamic time warping in linear time and space," *Intell. Data Anal.*, vol. 11, no. 5, pp. 561–580, Oct. 2007.
- [31] F. Zhou and F. Torre, "Canonical time warping for alignment of human behavior," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 22, 2009, pp. 2286–2294.

⁴The feasible set in POW is $\Pi_s(\mathbf{u}_{n_u}, \mathbf{u}_{n_v}) = \left\{ \mathbf{T} \in \mathbb{R}_+^{n_u \times n_v} \mid \mathbf{T} \mathbf{1}_{n_v} \leq \mathbf{u}_{n_u}, \mathbf{T}^\top \mathbf{1}_{n_u} \leq \mathbf{u}_{n_v}, \mathbf{1}_{n_u}^\top \mathbf{T} \mathbf{1}_{n_v} = s \right\}$, which reduces to that in (19) when $s = 1$.

- [32] M. Cuturi and M. Blondel, "Soft-DTW: A differentiable loss function for time-series," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 894–903.
- [33] T. Doan and T. Atsuhiro, "Deep multi-view learning from sequential data without correspondence," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2019, pp. 1–8.
- [34] W. Choi, J. Cho, S. Lee, and Y. Jung, "Fast constrained dynamic time warping for similarity measure of time series data," *IEEE Access*, vol. 8, pp. 222841–222858, 2020.
- [35] D. P. Tung and A. Takasu, "Deep multiview learning from sequentially unaligned data," *IEEE Access*, vol. 8, pp. 217928–217946, 2020.
- [36] M. Dvornik, I. Hadji, K. G. Derpanis, A. Garg, and A. Jepson, "Drop-DTW: Aligning common signal between sequences while dropping outliers," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 13782–13793.
- [37] M. Herrmann and G. I. Webb, "Amercing: An intuitive and effective constraint for dynamic time warping," *Pattern Recognit.*, vol. 137, May 2023, Art. no. 109333.
- [38] L. Qiu, C. Qiu, and C. Song, "ESDTW: Extrema-based shape dynamic time warping," *Expert Syst. Appl.*, vol. 239, Apr. 2024, Art. no. 122432.
- [39] Q. Cai, L. Chen, J. Shao, and L. Chen, "AdaPDTW: An efficient abstract-adaptive piecewise dynamic time warping for time series classification," *IEEE Access*, vol. 13, pp. 84188–84201, 2025.
- [40] F. Albrechtsen, "Statistical texture measures computed from gray level cooccurrence matrices," Dept. of Informatics, University of Oslo, Oslo, Norway, Tech. Rep., 2008.
- [41] T. Le, M. Yamada, K. Fukumizu, and M. Cuturi, "Tree-sliced variants of Wasserstein distances," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 12283–12294.
- [42] M. Yamada, Y. Takezawa, R. Sato, H. Bao, Z. Kozareva, and S. Ravi, "Approximating 1-Wasserstein distance with trees," *Trans. Mach. Learn. Res.*, Sep. 2022.
- [43] C. Semple and M. Steel, *Oxford Lecture Series in Mathematics and Its Applications*, vol. 24. London, U.K.: Oxford Univ. Press, 2003.
- [44] H. V. Tran, H. T. Pham, T. T. Huu, M.-K. Nguyen-Nhat, T. Chu, T. Le, and T. M. Nguyen, "Tree-sliced Wasserstein distance: A geometric perspective," in *Proc. 42nd Int. Conf. Mach. Learn.*, 2025, pp. 59969–60000.
- [45] T. Tran, H. V. Tran, T. Chu, H. T. Pham, L. Ghaoui, T. Le, and T. M. Nguyen, "Tree-sliced Wasserstein distance with nonlinear projection," in *Proc. 42nd Int. Conf. Mach. Learn.*, 2025, pp. 60001–60033.
- [46] H. V. Tran, T. Chu, M.-K. Nguyen-Nhat, H. T. Pham, T. Le, and T. M. Nguyen, "Spherical tree-sliced Wasserstein distance," in *Proc. 13th Int. Conf. Learn. Represent.*, 2025, pp. 36186–36218.
- [47] H. V. Tran, T. Chu, M.-K. Nguyen-Nhat, H. T. Pham, T. Le, and T. M. Nguyen, "Distance-based tree-sliced Wasserstein distance," in *Proc. 13th Int. Conf. Learn. Represent.*, 2025.
- [48] T. F. Gonzalez, "Clustering to minimize the maximum intercluster distance," *Theor. Comput. Sci.*, vol. 38, pp. 293–306, May 1985.
- [49] T. Feder and D. Greene, "Optimal algorithms for approximate clustering," in *Proc. 20th Annu. ACM Symp. Theory Comput.*, Jan. 1988, pp. 434–444.
- [50] T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms*. Cambridge, MA, USA: MIT Press, 2022.
- [51] C. Semple and M. Steel, *Phylogenetics*, vol. 24. London, U.K.: Oxford Univ. Press, 2003.
- [52] Z. Ghorrati, A. Esmaeli, and T. Eric Matson, "A multi-section hierarchical deep neural network model for time series classification: Applied to wearable sensor-based human activity recognition," *IEEE Access*, vol. 12, pp. 137851–137869, 2024.
- [53] A. R. Pitombeira-Neto, D. S. D. França, L. A. Cruz, T. L. C. D. Silva, and J. F. A. D. Macedo, "An ensemble Bayesian dynamic linear model for human activity recognition," *IEEE Access*, vol. 13, pp. 30316–30333, 2025.
- [54] G. P. C. P. Da Luz, D. H. P. Soto, O. O. Napoli, A. Rocha, L. Boccato, and E. Borin, "Benchmarking encoders and self-supervised learning for smartphone-based human activity recognition," *IEEE Access*, vol. 14, pp. 37451–37475, 2026.
- [55] D. Guijo-Rubio, M. Middlehurst, G. Arcencio, D. F. Silva, and A. Bagnall, "Unsupervised feature based algorithms for time series extrinsic regression," *Data Mining Knowl. Discovery*, vol. 38, no. 4, pp. 2141–2185, Jul. 2024.
- [56] T. de Waele, J. Fontaine, E. de Poorter, and A. Shahid, "Generalized ECG heartbeat classification using time-series transformers," *IEEE Access*, vol. 14, pp. 7699–7714, 2026.
- [57] T. Kim, M. Kang, J. Lee, J.-H. Bae, and Y. Sung, "DF-TAC arc Net: Domain fusion-based DC arc-fault classification using temporal adaptive method for distributed energy resources," *IEEE Access*, vol. 13, pp. 199216–199226, 2025.
- [58] T. Bohne, A.-K. P. Windler, and M. Atzmueller, "A domain-agnostic neuro-symbolic architecture for multimodal human-in-the-loop anomaly detection and complex fault diagnosis," *IEEE Access*, vol. 13, pp. 210201–210236, 2025.
- [59] Z. Xing, J. Pei, and E. Keogh, "A brief survey on sequence classification," *ACM SIGKDD Explorations Newslett.*, vol. 12, no. 1, pp. 40–48, Nov. 2010.
- [60] J. Lines and A. Bagnall, "Time series classification with ensembles of elastic distance measures," *Data Mining Knowl. Discovery*, vol. 29, no. 3, pp. 565–592, May 2015.
- [61] M. Blondel, A. Mensch, and J.-P. Vert, "Differentiable divergences between time series," in *Proc. Int. Conf. Artif. Intell. Statist.*, 2021, pp. 3853–3861.
- [62] Y. Ghahremani and B. Amiri, "Time series overlapping clustering based on link community detection," *IEEE Access*, vol. 12, pp. 41102–41124, 2024.
- [63] Z. Qi, H. Zhiyi, and W. Shaoren, "Interrupted sampling repeater jamming recognition and suppression method based on DBSCAN algorithm," *IEEE Access*, vol. 12, pp. 62597–62608, 2024.
- [64] R. Sorostinean, Z. Burghilea, and A. Gellert, "Anomaly detection in smart industrial machinery through hidden Markov models and autoencoders," *IEEE Access*, vol. 12, pp. 69217–69228, 2024.
- [65] P. Ivanov, M. Shtark, A. Kozhevnikov, M. Golyadkin, D. Botov, and I. Makarov, "SensorDBSCAN: Semi-supervised active learning powered method for anomaly detection and diagnosis," *IEEE Access*, vol. 13, pp. 25186–25197, 2025.
- [66] M. Gutoski, A. E. Lazzaretti, and H. S. Lopes, "Unsupervised open-world human action recognition," *Pattern Anal. Appl.*, vol. 26, no. 4, pp. 1753–1770, Nov. 2023.
- [67] N. Sedaghati, S. Ardebili, and A. Ghaffari, "Application of human activity/action recognition: A review," *Multimedia Tools Appl.*, vol. 84, no. 28, pp. 33475–33504, Jan. 2025.
- [68] L. Kaufman and P. J. Rousseeuw, *Finding Groups in Data: An Introduction to Cluster Analysis*. Hoboken, NJ, USA: Wiley, 2009.
- [69] J. Liu, Q. Li, H. Yang, Y. Han, S. Jiang, and W. Chen, "Sequence fault diagnosis for PEMFC water management subsystem using deep learning with t-SNE," *IEEE Access*, vol. 7, pp. 92009–92019, 2019.
- [70] K. Y. Wong and F.-L. Chung, "Visualizing time series data with temporal matching based t-SNE," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2019, pp. 1–8.
- [71] P. Soroosh, O. S. Kayhan, and J. C. van Gemert, "T-EVA: Time-efficient t-SNE video annotation," in *Proc. Int. Conf. Pattern Recognit.* Cham, Switzerland: Springer, 2021, pp. 153–169.
- [72] H. T. L. Huong, T. D. Cong, L. V. Vu, and L. K. Giang, "Latent pattern recognition in GNSS-based SHM using t-SNE and adaptive time-series modeling," *J. Civil Struct. Health Monitor.*, vol. 15, no. 8, pp. 3743–3766, Dec. 2025.
- [73] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 2579–2605, 2008.
- [74] E. H. Geoffrey and S. Roweis, "Stochastic neighbor embedding," in *Proc. Adv. Neural Inf. Process. Syst.*, 2002, pp. 833–840.
- [75] V. Max and M. Carreira-Perpinan, "Entropic affinities: Properties and efficient numerical computation," in *Proc. Int. Conf. Mach. Learn.*, 2013, pp. 477–485.
- [76] H. A. Dau, A. Bagnall, K. Kamgar, C. M. Yeh, Y. Zhu, S. Gharghabi, C. A. Ratanamahatana, and E. Keogh, "The UCR time series archive," *IEEE/CAA J. Autom. Sinica*, vol. 6, no. 6, pp. 1293–1305, Nov. 2019.
- [77] L. Gorelick, M. Blank, E. Shechtman, M. Irani, and R. Basri, "Actions as space-time shapes," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 29, no. 12, pp. 2247–2253, Dec. 2007.
- [78] W. Li, Z. Zhang, and Z. Liu, "Action recognition based on a bag of 3D points," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2010, pp. 9–14.
- [79] K. Bache and M. Lichman, "UCI machine learning repository," School Inf. Comput. Sci., University of California, Irvine, CA, USA, Tech. Rep., 2013.
- [80] D. Cai, X. He, and J. Han, "Document clustering using locality preserving indexing," *IEEE Trans. Knowl. Data Eng.*, vol. 17, no. 12, pp. 1624–1637, Dec. 2005.

- [81] L. Lovasz, *Matching Theory (North-Holland Mathematics Studies)*. Amsterdam, The Netherlands: Elsevier, 1986.
- [82] J. Venna and S. Kaski, "Local multidimensional scaling with controlled tradeoff between trustworthiness and continuity," in *Proc. 5th Workshop Self-Organizing Maps*, 2005, pp. 695–702.
- [83] L. Van Der Maaten, E. O. Postma, and H. J. Van Den Herik, "Dimensionality reduction: A comparative review," *J. Mach. Learn. Res.*, vol. 10, nos. 66–71, p. 13, 2009.
- [84] R. Gove, L. Cadalzo, N. Leiby, J. M. Singer, and A. Zaitzeff, "New guidance for using t-SNE: Alternative defaults, hyperparameter selection automation, and comparative evaluation," *Vis. Informat.*, vol. 6, no. 2, pp. 87–97, Jun. 2022.
- [85] L. Chen and A. Buja, "Local multidimensional scaling for nonlinear dimension reduction, graph drawing, and proximity analysis," *J. Amer. Stat. Assoc.*, vol. 104, no. 485, pp. 209–219, Mar. 2009.
- [86] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, Jun. 1947.
- [87] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *J. Mach. Learn. Res.*, vol. 7, pp. 1–30, May 2006.
- [88] A. P. Ruiz, M. Flynn, J. Large, M. Middlehurst, and A. Bagnall, "The great multivariate time series classification bake off: A review and experimental evaluation of recent algorithmic advances," *Data Mining Knowl. Discovery*, vol. 35, no. 2, pp. 401–449, Mar. 2021.
- [89] X. Ao, Y. Gong, and A. He, "A review of time series prediction models based on deep learning," *IEEE Access*, vol. 13, pp. 153696–153712, 2025.
- [90] U. A. Usmani, I. A. Aziz, J. Jaafar, and J. Watada, "Deep learning for anomaly detection in time-series data: An analysis of techniques, review of applications, and guidelines for future research," *IEEE Access*, vol. 12, pp. 174564–174590, 2024.
- [91] T. Doan and A. Takasu, "Kernel clustering with sigmoid regularization for efficient segmentation of sequential data," *IEEE Access*, vol. 10, pp. 62848–62862, 2022.



TUNG DOAN received the B.S. degree in computer engineering from Hanoi University of Science and Technology (HUST), in 2014, and the Ph.D. degree from the National Institute of Informatics, Japan, in 2021. He is currently a Lecturer with the Department of Computer Engineering, School of Information and Communication Technology, HUST. His current research interests include machine learning, optimal transport, and their applications on images and sequential data.



HOANG LE is currently pursuing the B.S. degree in computer science with Hanoi University of Science and Technology (HUST), Hanoi, Vietnam. He is currently a Research Assistant with the Machine Learning Laboratory, International Research Center for Artificial Intelligence (BKAI), HUST. His research interests include machine learning and optimal transport for sequential data.



HONG VUONG received the B.S. degree from VNU University of Engineering and Technology (VNU-UET), Vietnam, in 2016, and the Ph.D. degree from the National Institute of Informatics, Japan, in 2025. She is currently a Lecturer with the Faculty of Information Technology, VNU-UET. Her research interests include deep learning, multi-view representation learning, sensor data alignment, and sensor-based human activity recognition.

...